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Inference for block progressive censored competing risks data from an inverted exponentiated exponential model

Liang Wang[1] | **Shuo-Jye Wu**[2] | **Huizhong Lin**[1] | **Yogesh Mani Tripathi**[3]

[1]School of Mathematics, Yunnan Normal University, Kunming, PR China

[2]Department of Statistics, Tamkang University, Tamsui Taipei, Taiwan ROC

[3]Department of Mathematics, Indian Institute of Technology Patna, Patna, India

Correspondence
Liang Wang, School of Mathematics, Yunnan Normal University, No. 768, Juxian Rd, Chenggong District, Kunming 650500, PR China.
Email: liang610112@163.com

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Abstract

In this paper, reliability estimation for a competing risks model is discussed under a block progressive censoring scheme, which improves experimental efficiency through testing items under different testing facilities. When the lifetime of units follows an inverted exponentiated exponential distribution (IEED) and taking difference in testing facilities into account, various approaches are established for estimating unknown parameters, reliability performances and the differences in different testing facilities. Maximum likelihood estimators of IEED competing risks parameters together with existence and uniqueness are established, and the reliability performances and the difference in different testing facilities are also obtained in consequence. In addition, a hierarchical Bayes approach is proposed and the Metropolis-Hastings sampling algorithm is constructed for complex posterior computation. Finally, extensive simulation studies and a real data analysis are carried out to elaborate the performance of the methods, and the numerical results show that the proposed hierarchical Bayes model outperforms than classical likelihood method under block progressive censoring.

KEYWORDS

block progressive censoring, competing risks model, differences in testing facilities, hierarchical Bayesian model, inverted exponentiated exponential distribution, likelihood estimation

1 | INTRODUCTION

Censoring is a commonly used phenomenon in lifetime experiment when the cost and time constraints are considered where traditional complete experiments may have a long test cycle. In practice, different censoring schemes (CSs) are conducted for improving efficiency and saving cost. In practice, progressive Type-II censoring is the most popular one due to its flexibility and efficiency which can be conducted as follows. Assume that n identical units are put in a test with pre-fixed CS $R = (r_1, r_2, \ldots, r_m)$ with integer $m < n$. When i-th ($i = 1, 2, \ldots, m$) failure time $X_{i:m:n}$ occurred, r_i units are randomly removed from remaining surviving items and the then experiment stops. It is noted that traditional complete and Type-II censoring are two special cases of the progressively Type-II censoring. Under different censoring scenarios, extensive studies have been discussed by many authors. See, for example, some recent contributions of Ahmad et al.,[1] Altindag and Aydogdu,[2] Kohansal and Shoaee,[3] Laumen and Cramer,[4] Maurya and Tripathi,[5] Mondal and Kundu[6] Yan et al.[7] among others. For more details, one could refer to monographes of Lawless[8] and Balakrishnan and Cramer.[9]

Recently, in order to improve test efficiency, a new flexible procedure namely block censoring scheme (BCS) was proposed by Ahmadi et al.,[10] and the associated procedure is performed as follows. Suppose n identical units are divided

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into k groups with $n_i (i = 1, 2, \ldots, k)$ units in i-th group. Then the k groups are tested with k facilities under Type-II censoring. From applicative perspectives, one potential benefit of BCS is that, sometimes it is hard for an inspector to test all units simultaneously, then BCS provides a flexible way in real life experiments. In addition, due to some practical restrictions, it is also not available the necessary equipment for putting all the units on the testing simultaneously. Therefore, the BCS may provide a more effective way to test units at multiple groups. After proposed by Ahmadi et al.[10] where excellent studies are discussion on exponential block censored data, Zhu[11] further developed classical and Bayesian inference for Weibull model under block censoring. Then, there is rare publication on BCS according to our best acknowledge.

In real-world analysis, it is common to encounter multiple causes of failure, and the unit's lifespan is determined by the earliest occurrence among all causes. In literature, these causes of failure are called competing risks, and associated observations include failure times and cause indicator. Clearly, such failure causes contain potential information as same as failure times in competing risk model, and ignoring such information sometimes may yield inappropriate even misleading inferential results. Extensive studies have been discussed for competing risks model under various approaches, see, for example, the works of Liu et al.,[12] Wang et al.,[13] Wu et al.[14] One could refer to monograph of Crowder[15] for details.

Let T be a random variable from an inverted exponentiated exponential distribution (IEED), cumulative distribution function (CDF) and probability density function (PDF) of T are respectively given by

$$F(t;\theta,\lambda) = 1 - [1 - e^{-\lambda/t}]^{\theta} \text{ and } f(t;\theta,\lambda) = \frac{\theta\lambda}{t^2}\frac{e^{-\lambda/t}}{[1-e^{-\lambda/t}]^{\theta-1}}, \quad t > 0, \tag{1}$$

where $\theta > 0$ and $\lambda > 0$ are shape and scale parameters, respectively. Correspondingly, survival function (SF) and hazard rate function (HRF) of IEED can be expressed respectively as

$$S(t) = [1 - e^{-\lambda/t}]^{\theta} \quad \text{and} \quad h(t) = \frac{\theta\lambda}{t^2}\frac{e^{-\lambda/t}}{1-e^{-\lambda/t}}. \tag{2}$$

In addition, since mean time to failure of IEED has no closed form, a reliability index called median time to failure (MTF) is given by

$$\text{MTF} = \frac{-\lambda}{\ln[1-(0.5)^{1/\theta}]}. \tag{3}$$

Hereafter, we use IEED(θ,λ) denoting the IEED with parameters θ and λ. The IEED was first proposed by Abouammoh and Alshingiti,[16] and has been discussed by many authors (e.g., Dey et al.,[17] Dey and Nassar,[18] Dube et al.,[19] Kinaci et al.,[20] Krishna et al.,[21] Panahi[22]). It's concluded that the IEED can be considered as an useful alternative due to its flexible data fitting ability as gamma, Weibull, generalized exponential and inverted exponential distributions, whereas the IEED also appears attractive applications in the fields of accelerated life testing, horse racing, supermarket queues, ocean currents, wind speed among others.

In Ahmadi et al.[10]'s BCS, lifetime experiment is conducted under multiple groups with Type-II censoring. However, such scenario is time and cost consuming due to high reliable and long life-cycle of modern products. Moreover, experiment sometimes needs to meet some practical requirements for conducting a more flexible test. For example, some alive units under studies are need to be removed for other tests that is particularly useful when the units are very expensive. Therefore, it is essential to carry out BCS under more general scenarios from practical perspective. In addition, following Ahmadi et al.,[10] it is noted that the testing facilities are treated as same equipments under BCS scenario, but failure information collected from different testing facilities are not exactly identical under such supposed identical testing facilities from practical perspectives. Any influencing factors such as facilities' operating environment, personnel testing specification, data collection accuracy may cause noneligible difference among the data from different testing facilities. Therefore, it is also reasonable to treat failure data as non-identical records and the associated difference in different facilities (DDF) as nonnegligible in BCS, whereas ignoring the DDF in inference sometimes may result improper even misleading analytical results. Motivated by the aforementioned reasons and due to the potential applications of IEED, this paper is devoted to discuss reliability estimation of competing risks model under a proposed block progressive censoring scheme (BPCS).

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TABLE 1 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_2, \lambda) = (0.25, 0.55, 1)$ with $k = 2$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.0910[0.0082]	0.9145[0.7212]	0.0012[0.0005]	0.9384[0.1423]
		θ_{12}	0.2975[0.0885]	0.9017[1.2758]	0.0021[0.0002]	0.9186[0.5471]
		θ_{21}	0.1188[0.0141]	0.8926[1.7499]	0.0014[0.0006]	0.9088[0.5234]
		θ_{22}	0.2636[0.0695]	0.8863[2.5913]	0.0034[0.0003]	0.9045[0.5913]
		θ_1	0.3348[0.2307]	0.9187[0.6021]	0.0038[0.0002]	0.9187[0.4563]
		θ_2	0.8742[1.8847]	0.9023[1.3064]	0.0031[0.0002]	0.9259[0.6348]
		λ	3.6214[13.115]	0.8994[1.3742]	1.1054[1.4578]	0.9218[0.6018]
		$S(t)$	0.2904[0.0845]	0.9356[0.0164]	0.6928[0.4800]	0.9448[0.0032]
		$h(t)$	0.3866[0.1544]	0.9311[0.0646]	0.5247[0.3418]	0.9459[0.0073]
		Md(t)	6.42E2[4.10E4]	0.9345[0.0652]	22.9217[5.2E2]	0.9473[0.0074]
	40	θ_{11}	0.0658[0.0043]	0.9543[0.1844]	0.0008[0.0001]	0.9683[0.1621]
		θ_{12}	0.2213[0.0490]	0.9436[0.3018]	0.0009[0.0002]	0.9561[0.2192]
		θ_{21}	0.0862[0.0074]	0.9365[0.3372]	0.0009[0.0003]	0.9443[0.2294]
		θ_{22}	0.1560[0.0243]	0.9247[0.4990]	0.0011[0.0001]	0.9373[0.3145]
		θ_1	0.1831[0.0405]	0.9589[0.1541]	0.0012[0.0001]	0.9642[0.1238]
		θ_2	0.2151[0.0929]	0.9401[0.2682]	0.0011[0.0001]	0.9567[0.1543]
		λ	3.0187[9.1128]	0.9052[0.9441]	0.5613[0.7439]	0.9230[0.4806]
		$S(t)$	0.2959[0.0876]	0.9478[0.0093]	0.2846[0.3165]	0.9525[0.0041]
		$h(t)$	0.4195[0.1769]	0.9504[0.0308]	0.3044[0.1607]	0.9641[0.0083]
		Md(t)	1.31E2[1.52E4]	0.9496[0.0307]	10.6483[2.80E2]	0.9594[0.0067]
100	50	θ_{11}	0.1278[0.0163]	0.9641[0.1393]	0.0035[0.0017]	0.9213[0.1649]
		θ_{12}	0.3425[0.1173]	0.9563[0.2177]	0.0034[0.0014]	0.8941[0.6256]
		θ_{21}	0.1977[0.0391]	0.9418[0.2379]	0.0027[0.0016]	0.8848[0.6487]
		θ_{22}	0.3769[0.1420]	0.9377[0.3436]	0.0043[0.0011]	0.8857[0.6324]
		θ_1	0.2860[0.0825]	0.9635[0.1152]	0.0079[0.0010]	0.9074[0.5583]
		θ_2	0.3010[0.0940]	0.9534[0.1889]	0.0061[0.0013]	0.9135[0.7121]
		λ	2.6449[6.9955]	0.9128[0.8122]	2.4371[1.7306]	0.9107[0.7145]
		$S(t)$	0.3003[0.0902]	0.9473[0.0101]	0.8347[0.5384]	0.9216[0.0068]
		$h(t)$	0.4415[0.1950]	0.9586[0.0296]	0.6345[0.5843]	0.9248[0.0091]
		Md(t)	1.51E2[2.23E4]	0.9539[0.0284]	31.4527[6.1E2]	0.9273[0.0090]
	70	θ_{11}	0.0038[0.0002]	0.9758[0.0473]	0.0013[0.0009]	0.9354[0.1528]
		θ_{12}	0.0423[0.0017]	0.9687[0.0750]	0.0011[0.0006]	0.9173[0.5572]
		θ_{21}	0.0568[0.0032]	0.9542[0.0718]	0.0017[0.0009]	0.9067[0.5183]
		θ_{22}	0.0926[0.0085]	0.9465[0.1023]	0.0022[0.0005]	0.9016[0.4371]
		θ_1	0.1959[0.0430]	0.9784[0.0398]	0.0045[0.0004]	0.9183[0.4321]
		θ_2	0.1607[0.0377]	0.9721[0.0584]	0.0030[0.0007]	0.9281[0.3409]
		λ	2.4460[5.9832]	0.9294[0.7006]	1.2473[0.9438]	0.9346[0.5317]
		$S(t)$	0.2935[0.0861]	0.9613[0.0050]	0.6249[0.4147]	0.9355[0.0027]
		$h(t)$	0.4146[0.1725]	0.9584[0.0137]	0.5128[0.2158]	0.9318[0.0050]
		Md(t)	1.17E2[1.36E4]	0.9592[0.0144]	11.4789[2.90E2]	0.9324[0.0041]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

When lifetimes of competing risks of units follow IEEDs and progressive censoring scheme is implemented in block testing scenario, estimation of model parameters and reliability performances are discussed when the DDF is nonnegligible being taken into account. To our best knowledge, there is no publication on BPCS with competing risks data in literature.

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TABLE 2 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.2, 0.5, 1.5)$ with $k = 2$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.1101[0.0121]	0.9002[0.5453]	0.0021[0.0011]	0.9211[0.1108]
		θ_{12}	0.2835[0.0499]	0.8954[1.0654]	0.0032[0.0014]	0.9309[0.6107]
		θ_{21}	0.1402[0.0196]	0.8873[1.3258]	0.0043[0.0017]	0.8973[0.5913]
		θ_{22}	0.4207[0.0487]	0.8831[2.1682]	0.0045[0.0013]	0.8941[0.6547]
		θ_1	0.1410[0.0656]	0.9128[0.4647]	0.0056[0.0009]	0.9087[0.5391]
		θ_2	0.3540[0.3307]	0.8989[1.0294]	0.0054[0.0010]	0.9148[0.7168]
		λ	3.6311[13.1851]	0.8895[1.5001]	1.9431[2.5784]	0.9162[0.8589]
		$S(t)$	0.1564[0.0245]	0.9241[0.0063]	0.7630[0.5361]	0.9381[0.0057]
		$h(t)$	0.2712[0.0749]	0.9247[0.0299]	0.6473[0.4127]	0.9373[0.0084]
		Md(t)	7.80E3[6.01E4]	0.9242[0.0301]	24.7518[5.41E2]	0.9345[0.0096]
	40	θ_{11}	0.1052[0.0110]	0.9125[0.1518]	0.0011[0.0004]	0.9367[0.0741]
		θ_{12}	0.2708[0.0733]	0.9154[0.2729]	0.0016[0.0005]	0.9458[0.4863]
		θ_{21}	0.1097[0.0120]	0.9073[0.2747]	0.0017[0.0009]	0.9188[0.3524]
		θ_{22}	0.3602[0.1297]	0.9086[0.4188]	0.0020[0.0006]	0.9173[0.4712]
		θ_1	0.0841[0.0099]	0.9268[0.1309]	0.0028[0.0003]	0.9246[0.2155]
		θ_2	0.2047[0.0577]	0.9287[0.2238]	0.0021[0.0002]	0.9273[0.4871]
		λ	2.7101[7.3449]	0.9135[0.9738]	0.6849[1.0874]	0.9356[0.3458]
		$S(t)$	0.1538[0.0236]	0.9361[0.0074]	0.4127[0.2507]	0.9424[0.0027]
		$h(t)$	0.2666[0.0714]	0.9354[0.0259]	0.2843[0.1876]	0.9475[0.0037]
		Md(t)	1.23E2[5.81E3]	0.9352[0.0259]	10.4716[2.57E2]	0.9487[0.0043]
100	50	θ_{11}	0.0837[0.0070]	0.9247[0.0713]	0.0015[0.0008]	0.9368[0.0634]
		θ_{12}	0.1481[0.0219]	0.9283[0.1290]	0.0020[0.0009]	0.9473[0.5361]
		θ_{21}	0.0317[0.0010]	0.9141[0.1244]	0.0027[0.0008]	0.9137[0.4173]
		θ_{22}	0.1938[0.0375]	0.9152[0.1909]	0.0018[0.0005]	0.9134[0.3617]
		θ_1	0.1236[0.0163]	0.9370[0.0617]	0.0041[0.0002]	0.9292[0.3178]
		θ_2	0.2791[0.0866]	0.9348[0.1023]	0.0038[0.0003]	0.9356[0.4632]
		λ	2.5767[6.6398]	0.9219[0.8620]	0.7592[1.0814]	0.9341[0.6471]
		$S(t)$	0.1567[0.0245]	0.9475[0.0044]	0.5483[0.3147]	0.9490[0.0031]
		$h(t)$	0.2799[0.0784]	0.9465[0.0145]	0.2184[0.1247]	0.9493[0.0046]
		Md(t)	1.52E2[1.24E3]	0.9462[0.0144]	15.4278[4.12E2]	0.9489[0.0043]
	70	θ_{11}	0.0749[0.0132]	0.9352[0.0298]	0.0005[0.0003]	0.9493[0.0421]
		θ_{12}	0.1408[0.0161]	0.9346[0.0542]	0.0011[0.0003]	0.9581[0.0483]
		θ_{21}	0.1037[0.0008]	0.9283[0.0473]	0.0014[0.0004]	0.9264[0.0367]
		θ_{22}	0.3286[0.0437]	0.9275[0.0707]	0.0006[0.0002]	0.9287[0.0316]
		θ_1	0.0521[0.0134]	0.9480[0.0262]	0.0015[0.0001]	0.9393[0.1784]
		θ_2	0.1907[0.0539]	0.9491[0.0394]	0.0014[0.0001]	0.9428[0.2245]
		λ	0.9886[0.9773]	0.9335[0.6395]	0.0012[0.0001]	0.9462[0.2807]
		$S(t)$	0.1487[0.0221]	0.9553[0.0031]	0.3217[0.0034]	0.9508[0.0014]
		$h(t)$	0.2634[0.0716]	0.9576[0.0173]	0.2164[0.0437]	0.9581[0.0031]
		Md(t)	1.14E2[3.29E2]	0.9587[0.0094]	8.1045[1.09E2]	0.9593[0.0021]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

The article is organized as follows. In Section 2, model description and likelihood function is provided. Section 3 establishes classical estimation of unknown parameters and reliability performances. Bayesian approach is presented in Section 4. Numerical studies and a real-life example is conducted in Section 5. Finally, Section 6 gives some brief concluding remarks.

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TABLE 3 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.25, 0.55, 1)$ with $k = 3$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.0954[0.0096]	0.9121[0.7539]	0.0017[0.0009]	0.9296[0.1538]
		θ_{12}	0.3031[0.0927]	0.8963[1.4263]	0.0035[0.0006]	0.9191[0.5712]
		θ_{21}	0.1247[0.0183]	0.8904[1.7920]	0.0023[0.0008]	0.9032[0.5467]
		θ_{22}	0.2676[0.0728]	0.8841[2.6259]	0.0047[0.0005]	0.9017[0.6073]
		θ_{31}	0.1423[0.0145]	0.8911[1.3587]	0.0036[0.0007]	0.9078[0.5534]
		θ_{32}	0.2548[0.0613]	0.8839[1.3846]	0.0041[0.0008]	0.9025[0.5240]
		θ_1	0.3463[0.2461]	0.9138[0.6348]	0.0043[0.0005]	0.9155[0.4681]
		θ_2	0.3552[0.3348]	0.8924[1.0081]	0.0057[0.0013]	0.9122[0.7058]
		λ	3.7482[13.3746]	0.8843[1.5139]	1.9680[2.7342]	0.9051[0.8769]
		$S(t)$	0.1724[0.0273]	0.9211[0.0086]	0.7891[0.5849]	0.9360[0.0062]
		$h(t)$	0.2867[0.0785]	0.9172[0.0314]	0.6698[0.4356]	0.9265[0.0091]
		Md(t)	7.91E2[6.23E3]	0.9195[0.0359]	25.6143[5.52E2]	0.9287[0.0114]
	40	θ_{11}	0.1094[0.0118]	0.9063[0.1672]	0.0020[0.0007]	0.9241[0.0752]
		θ_{12}	0.2862[0.0857]	0.9106[0.2810]	0.0022[0.0006]	0.9418[0.4924]
		θ_{21}	0.1143[0.0169]	0.9126[0.2893]	0.0021[0.0012]	0.9069[0.3681]
		θ_{22}	0.3693[0.1317]	0.9040[0.4198]	0.0041[0.0011]	0.9127[0.4825]
		θ_{31}	0.1254[0.0183]	0.9136[0.3152]	0.0026[0.0008]	0.9083[0.2467]
		θ_{32}	0.2576[0.0957]	0.9085[0.4103]	0.0030[0.0010]	0.9096[0.3168]
		θ_1	0.0863[0.0112]	0.9150[0.1346]	0.0034[0.0007]	0.9211[0.2189]
		θ_2	0.2121[0.0577]	0.9230[0.2518]	0.0027[0.0006]	0.9142[0.4987]
		λ	2.9438[7.5683]	0.9083[0.9869]	0.6888[1.1644]	0.9245[0.3621]
		$S(t)$	0.1672[0.0294]	0.9257[0.0082]	0.4249[0.2608]	0.9370[0.0038]
		$h(t)$	0.2741[0.0761]	0.9315[0.0279]	0.2928[0.1991]	0.9413[0.0041]
		Md(t)	1.3E2[5.80E3]	0.9243[0.0372]	11.4928[2.81E2]	0.9371[0.0069]
100	50	θ_{11}	0.0928[0.0085]	0.9152[0.0746]	0.0023[0.0011]	0.9247[0.0748]
		θ_{12}	0.1539[0.0341]	0.9165[0.1383]	0.0028[0.0011]	0.9417[0.5872]
		θ_{21}	0.0393[0.0018]	0.9141[0.1362]	0.0031[0.0009]	0.9042[0.4418]
		θ_{22}	0.1938[0.0429]	0.9093[0.2133]	0.0021[0.0007]	0.9052[0.4370]
		θ_{31}	0.0637[0.0067]	0.9042[0.1745]	0.0024[0.0008]	0.9073[0.4713]
		θ_{32}	0.1746[0.0382]	0.9081[0.1534]	0.0027[0.0008]	0.9028[0.4135]
		θ_1	0.1381[0.0219]	0.9246[0.0724]	0.0053[0.0005]	0.9261[0.3280]
		θ_2	0.2956[0.0866]	0.9259[0.1286]	0.0042[0.0007]	0.9289[0.2332]
		λ	2.8192[6.9461]	0.9167[0.8733]	0.7638[1.0975]	0.9293[0.6611]
		$S(t)$	0.1743[0.0356]	0.9396[0.0051]	0.5569[0.3638]	0.9345[0.0049]
		$h(t)$	0.2830[0.0827]	0.9279[0.0283]	0.2184[0.1368]	0.9476[0.0049]
		Md(t)	1.64E2[1.38E3]	0.9321[0.0268]	15.6138[4.12E2]	0.9392[0.0057]
	70	θ_{11}	0.0776[0.0194]	0.9234[0.0349]	0.0008[0.0005]	0.9367[0.0579]
		θ_{12}	0.1511[0.0243]	0.9265[0.0680]	0.0026[0.0007]	0.9479[0.0592]
		θ_{21}	0.1261[0.0015]	0.9197[0.0582]	0.0020[0.0006]	0.9238[0.0451]
		θ_{22}	0.3474[0.0553]	0.9216[0.0798]	0.0009[0.0004]	0.9267[0.0480]
		θ_{31}	0.1452[0.0029]	0.9128[0.0543]	0.0022[0.0008]	0.9233[0.0421]
		θ_{32}	0.2084[0.0165]	0.9123[0.0627]	0.0016[0.0009]	0.9247[0.0438]
		θ_1	0.0543[0.0197]	0.9364[0.0312]	0.0020[0.0004]	0.9388[0.1846]
		θ_2	0.2046[0.0573]	0.9385[0.0418]	0.0016[0.0004]	0.9399[0.2571]
		λ	0.9988[0.9845]	0.9310[0.6459]	0.0015[0.0004]	0.9382[0.2986]

(Continues)

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TABLE 3 (Continued)

n	*m*	parameters	MLE	ACI	Bayes	HPD
		$S(t)$	0.1551[0.0246]	0.9492[0.0048]	0.3357[0.0047]	0.9504[0.0016]
		$h(t)$	0.2729[0.0820]	0.9461[0.0205]	0.2376[0.0519]	0.9491[0.0041]
		Md(t)	1.22E2[3.30E3]	0.9436[0.0113]	8.2497[2.01E2]	0.9496[0.0034]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

2 | MODEL DESCRIPTION AND LIKELIHOOD FUNCTION

2.1 | Testing strategy and data description

For BPCS with n identical units, lifetime of units has independent failure of causes following IEED(θ_1, λ) and IEED(θ_2, λ), respectively. In test, n identical units are divided into k groups with sample size $n_i, i = 1, 2, \dots, k$ satisfying $\sum_{i=1}^{k} n_i = n$. For i-th testing group, prefixed integer m_i and CS $R_i = (r_{i1}, r_{i2}, \dots, r_{im_i})$ are given. Following the scenario of progressive Type-II censoring, failure times $(X_{i1}, X_{i2}, \dots, X_{im_i})$ are obtained under i-th group, and following observations are obtained as

$$\begin{array}{cccccc} group & data\ id & samples & & & \\ 1 & \mathcal{D}_1 & (x_{11}, \delta_{11}, r_{11}), (x_{12}, \delta_{12}, r_{12}), \dots, (x_{1m_1}, \delta_{1m_1}, r_{1m_1}) & & & \\ 2 & \mathcal{D}_2 & (x_{21}, \delta_{21}, r_{21}), (x_{22}, \delta_{22}, r_{22}), \dots, (x_{2m_2}, \delta_{2m_2}, r_{2m_2}) & & & \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ k & \mathcal{D}_k & (x_{k1}, \delta_{k1}, r_{k1}), (x_{k2}, \delta_{k2}, r_{k2}), \dots, (x_{km_k}, \delta_{km_k}, r_{km_k}) & & & \end{array} \tag{4}$$

where $\delta_{ij}, i = 1, 2, \dots, k; j = 1, 2, \dots, m_i$ denotes the observed failure cause for the j-th progressive censored data in i-th group, which is defined as

$$\delta_{ij} = \begin{cases} 1, & \text{failure due to cause one,} \\ 2, & \text{failure due to cause two.} \end{cases}$$

Under BPCS, since the DDF among test groups is nonnegligible, in order to estimate DDF, it is further assumed that the competing risk parameter θ_1 and θ_2 are different in k testing groups. Then the failure times of units from i-th facilities are characterized by IEED competing risks model with parameters θ_{i1}, θ_{i2} and λ, where λ are assumed to be same in different groups reflecting the common failure mechanism of units. Specifically, the failure of units in the i-th ($i = 1, 2, \dots, k$) group is caused by two competing risks, and the s-th ($s = 1, 2$) cause of failure follows IEED with PDF and CDF as

$$f(t; \theta_{is}, \lambda) = \frac{\theta_{is}\lambda}{t^2} e^{-\lambda/t} [1 - e^{-\lambda/t}]^{\theta_{is}-1}, t > 0. \tag{5}$$

and

$$F(t; \theta_{is}, \lambda) = 1 - [1 - e^{-\lambda/t}]^{\theta_{is}}, t > 0. \tag{6}$$

Further, the minimal lifetime of two competing risks under i-th group is distributed by IEED($\theta_{i1} + \theta_{i2}, \lambda$) with SF

$$S(t; \theta_{i1} + \theta_{i2}, \lambda) = [1 - e^{-\lambda/t}]^{\theta_{i1}+\theta_{i2}}. \tag{7}$$

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TABLE 4 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.2, 0.5, 1.5)$ with $k = 3$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.1213[0.0169]	0.8837[0.8346]	0.0045[0.0018]	0.9073[0.1682]
		θ_{12}	0.4163[0.0438]	0.8936[1.2574]	0.0051[0.0032]	0.9103[0.7841]
		θ_{21}	0.2135[0.0243]	0.8768[1.5286]	0.0051[0.0023]	0.8846[0.6245]
		θ_{22}	0.5319[0.0647]	0.8653[2.4718]	0.0053[0.0034]	0.8752[0.7186]
		θ_{31}	0.3461[0.0452]	0.8942[1.2746]	0.0047[0.0039]	0.9035[0.6317]
		θ_{32}	0.4518[0.0583]	0.8963[1.5278]	0.0046[0.0033]	0.9071[0.6843]
		θ_1	0.1673[0.0842]	0.9045[0.5328]	0.0064[0.0015]	0.9246[0.4135]
		θ_2	0.4271[0.4213]	0.9076[0.6217]	0.0061[0.0014]	0.9099[0.4762]
		λ	3.8245[14.6381]	0.8723[1.2048]	2.4631[3.1485]	0.9035[0.8924]
		$S(t)$	0.1830[0.0468]	0.9168[0.0072]	0.8124[0.5847]	0.9213[0.0068]
		$h(t)$	0.3246[0.0825]	0.9135[0.0358]	0.7241[0.4823]	0.9257[0.0093]
		Md(t)	8.42E2[6.31E4]	0.9145[0.0456]	31.5287[5.93E2]	0.9263[0.0061]
	40	θ_{11}	0.0932[0.0083]	0.9160[0.0837]	0.0017[0.0011]	0.9246[0.0723]
		θ_{12}	0.1539[0.0305]	0.9123[0.1458]	0.0029[0.0013]	0.9338[0.4586]
		θ_{21}	0.0458[0.0016]	0.9067[0.1354]	0.0036[0.0009]	0.9178[0.0568]
		θ_{22}	0.2014[0.0485]	0.9013[0.2089]	0.0024[0.0008]	0.9049[0.4251]
		θ_{31}	0.0684[0.0056]	0.9085[0.1235]	0.0027[0.0012]	0.9204[0.0432]
		θ_{32}	0.1765[0.0315]	0.9141[0.2348]	0.0025[0.0011]	0.9024[0.5037]
		θ_1	0.0927[0.0111]	0.9128[0.1542]	0.0037[0.0006]	0.9163[0.3276]
		θ_2	0.2256[0.0635]	0.9174[0.2473]	0.0034[0.0005]	0.9189[0.5563]
		λ	3.8724[8.7561]	0.9060[1.0548]	0.7246[1.2487]	0.9215[0.4658]
		$S(t)$	0.1639[0.0376]	0.9237[0.0083]	0.5236[0.3541]	0.9293[0.0036]
		$h(t)$	0.3542[0.0825]	0.9283[0.0256]	0.2836[0.1935]	0.9376[0.0042]
		Md(t)	1.34E2[5.91E3]	0.9237[0.0348]	12.5873[2.94E2]	0.9362[0.0051]
100	50	θ_{11}	0.0926[0.0074]	0.9231[0.0834]	0.0023[0.0009]	0.9352[0.0638]
		θ_{12}	0.1493[0.0256]	0.9236[0.1336]	0.0025[0.0009]	0.9486[0.5376]
		θ_{21}	0.0431[0.0035]	0.9038[0.1386]	0.0024[0.0008]	0.9125[0.4192]
		θ_{22}	0.1973[0.0388]	0.9167[0.2046]	0.0022[0.0007]	0.9121[0.3624]
		θ_{31}	0.0467[0.0031]	0.9036[0.0461]	0.0026[0.0008]	0.9174[0.4153]
		θ_{32}	0.1542[0.0287]	0.9045[0.1673]	0.0027[0.0007]	0.9152[0.3568]
		θ_1	0.1267[0.0174]	0.9245[0.0708]	0.0053[0.0005]	0.9231[0.3548]
		θ_2	0.3146[0.0935]	0.9235[0.1197]	0.0041[0.0007]	0.9288[0.5249]
		λ	3.1869[8.1547]	0.9126[0.8834]	0.8623[1.5931]	0.9275[0.7132]
		$S(t)$	0.1638[0.0366]	0.9317[0.0052]	0.5546[0.3245]	0.9347[0.0038]
		$h(t)$	0.2834[0.0792]	0.9368[0.0176]	0.2318[0.1453]	0.9398[0.0052]
		Md(t)	1.74E2[1.31E3]	0.9356[0.0168]	17.5179[4.32E2]	0.9368[0.0067]
	70	θ_{11}	0.0764[0.0143]	0.9327[0.0315]	0.0009[0.0006]	0.9377[0.0543]
		θ_{12}	0.1362[0.0189]	0.9283[0.0561]	0.0023[0.0006]	0.9443[0.0591]
		θ_{21}	0.1106[0.0013]	0.9234[0.0489]	0.0032[0.0006]	0.9214[0.0426]
		θ_{22}	0.3571[0.0503]	0.9139[0.0745]	0.0009[0.0005]	0.9186[0.0347]
		θ_{31}	0.1463[0.0061]	0.9258[0.0543]	0.0021[0.0004]	0.9233[0.0634]
		θ_{32}	0.3218[0.0153]	0.9191[0.0428]	0.0009[0.0005]	0.9148[0.0315]
		θ_1	0.0539[0.0215]	0.9367[0.0287]	0.0028[0.0003]	0.9392[0.0156]
		θ_2	0.2057[0.0638]	0.9351[0.0395]	0.0021[0.0004]	0.9386[0.3092]
		λ	1.0432[0.9871]	0.9238[0.6582]	0.0017[0.0004]	0.9375[0.2936]
		$S(t)$	0.1658[0.0269]	0.9451[0.0042]	0.3526[0.0044]	0.9482[0.0069]

(Continues)

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TABLE 4 (Continued)

n	*m*	parameters	MLE	ACI	Bayes	HPD
		$h(t)$	0.3182[0.0824]	0.9436[0.0186]	0.2641[0.0519]	0.9494[0.0053]
		Md(t)	1.10E2[3.24E2]	0.9474[0.0098]	8.4563[1.93E2]	0.9485[0.0034]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

TABLE 5 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_2, \lambda) = (0.15, 0.45, 0.5)$ with $k = 2$.

n	*m*	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.1456[0.0091]	0.9001[0.7863]	0.0022[0.0009]	0.9140[0.1793]
		θ_{12}	0.3814[0.0926]	0.9009[1.3162]	0.0040[0.0007]	0.8835[0.6593]
		θ_{21}	0.2063[0.0215]	0.8814[1.8671]	0.0023[0.0011]	0.8729[0.6629]
		θ_{22}	0.4451[0.0725]	0.8628[2.6318]	0.0049[0.0008]	0.8705[0.6573]
		θ_1	0.3639[0.2632]	0.9022[0.6643]	0.0046[0.0005]	0.8935[0.5682]
		θ_2	0.3923[1.9258]	0.8943[1.4562]	0.0045[0.0008]	0.9024[0.7384]
		λ	3.6827[13.9823]	0.8722[1.5628]	1.3528[1.6247]	0.9135[0.7462]
		$R(t)$	0.3861[0.0963]	0.9136[0.0191]	0.6751[0.5147]	0.9190[0.0082]
		$h(t)$	0.4619[0.1762]	0.9135[0.0687]	0.5247[0.3418]	0.9106[0.0114]
		Md(t)	3.91E3[5.03E5]	0.9204[0.0835]	35.6428[8.13E2]	0.9137[0.0108]
	40	θ_{11}	0.1454[0.0082]	0.9428[0.2165]	0.0017[0.0008]	0.9207[0.2038]
		θ_{12}	0.3541[0.0681]	0.9214[0.3482]	0.0021[0.0007]	0.9173[0.5890]
		θ_{21}	0.1049[0.0093]	0.9143[0.3857]	0.0018[0.0011]	0.9004[0.5413]
		θ_{22}	0.1786[0.0418]	0.9062[0.5782]	0.0030[0.0006]	0.9002[0.4205]
		θ_1	0.2024[0.0682]	0.9348[0.1869]	0.0023[0.0008]	0.9044[0.4743]
		θ_2	0.2947[0.1165]	0.9235[0.3471]	0.0039[0.0007]	0.9193[0.3725]
		λ	4.3419[10.4287]	0.8829[1.1074]	0.7048[0.8193]	0.9286[0.6284]
		$R(t)$	0.3486[0.0981]	0.9293[0.0204]	0.3671[0.4288]	0.9207[0.0038]
		$h(t)$	0.4654[0.1947]	0.9225[0.0783]	0.3923[0.2636]	0.9241[0.0085]
		Md(t)	1.74E3[1.85E4]	0.9170[0.0428]	12.1542[3.31E2]	0.9164[0.0073]
100	50	θ_{11}	0.1063[0.0242]	0.9521[0.1542]	0.0046[0.0031]	0.9261[0.1542]
		θ_{12}	0.3247[0.1583]	0.9417[0.2433]	0.0054[0.0025]	0.9078[0.5556]
		θ_{21}	0.1362[0.0452]	0.9328[0.2688]	0.0040[0.0027]	0.8984[0.5421]
		θ_{22}	0.2896[0.1528]	0.9467[0.3624]	0.0054[0.0026]	0.8991[0.6049]
		θ_1	0.3019[0.0968]	0.9549[0.1368]	0.0087[0.0015]	0.9103[0.4682]
		θ_2	0.3456[0.1122]	0.9419[0.1967]	0.0079[0.0029]	0.9046[0.6164]
		λ	2.8816[1.0483]	0.9003[0.8463]	2.7418[1.9251]	0.9047[0.7213]
		$R(t)$	0.3048[0.0974]	0.9317[0.0159]	0.8526[0.5832]	0.9352[0.0053]
		$h(t)$	0.4039[0.2046]	0.9424[0.0349]	0.6543[0.6014]	0.9315[0.0086]
		Md(t)	1.92E3[2.84E4]	0.9421[0.0365]	25.9875[6.21E2]	0.9142[0.0087]
	70	θ_{11}	0.1110[0.0123]	0.9612[0.0542]	0.0021[0.0014]	0.9566[0.1839]
		θ_{12}	0.3165[0.1001]	0.9511[0.1047]	0.0018[0.0009]	0.9352[0.2988]
		θ_{21}	0.1263[0.0159]	0.9475[0.0856]	0.0024[0.0017]	0.9215[0.2976]
		θ_{22}	0.3864[0.1493]	0.9306[0.1618]	0.0026[0.0009]	0.9197[0.4617]
		θ_1	0.1069[0.0117]	0.9628[0.0512]	0.0051[0.0008]	0.9428[0.1723]
		θ_2	0.3176[0.1031]	0.9648[0.0695]	0.0042[0.0011]	0.9395[0.2498]
		λ	3.6144[13.0639]	0.9150[0.8204]	1.5924[0.9827]	0.9265[0.5718]
		$R(t)$	0.4258[0.1813]	0.9142[0.0137]	0.6093[0.4458]	0.9327[0.0068]
		$h(t)$	0.4507[0.2032]	0.9466[0.0249]	0.5540[0.2674]	0.9471[0.0096]

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TABLE 5 (Continued)

n	*m*	parameters	MLE	ACI	Bayes	HPD
		Md(t)	4.72E2[3.41E3]	0.9425[0.0285]	10.6473[2.53E2]	0.9349[0.0096]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

TABLE 6 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.3, 0.6, 2)$ with $k = 2$.

n	*m*	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.2248[0.0275]	0.8953[0.5871]	0.0035[0.0019]	0.9146[0.1482]
		θ_{12}	0.3352[0.0618]	0.8847[1.0962]	0.0039[0.0027]	0.9247[0.6948]
		θ_{21}	0.1683[0.0291]	0.8802[1.3689]	0.0049[0.0022]	0.8846[0.6201]
		θ_{22}	0.4874[0.0625]	0.8789[2.4528]	0.0055[0.0020]	0.8832[0.6957]
		θ_1	0.1637[0.0711]	0.9073[0.4873]	0.0061[0.0013]	0.9082[0.4184]
		θ_2	0.3902[0.3725]	0.8791[1.1496]	0.0060[0.0018]	0.9056[0.7452]
		λ	3.9841[14.1581]	0.8822[1.5764]	1.9883[2.7148]	0.9036[0.8749]
		$R(t)$	0.1868[0.0542]	0.9184[0.0075]	0.7938[0.5582]	0.9193[0.0071]
		$h(t)$	0.3016[0.0861]	0.9168[0.0347]	0.6925[0.4715]	0.9258[0.0096]
		Md(t)	8.12E3[6.43E5]	0.9173[0.0581]	26.8792[5.71E2]	0.9269[0.0098]
	40	θ_{11}	0.1564[0.0166]	0.9014[0.1963]	0.0021[0.0008]	0.9258[0.0879]
		θ_{12}	0.3043[0.0486]	0.9028[0.2987]	0.0019[0.0007]	0.9378[0.5269]
		θ_{21}	0.1241[0.0173]	0.9015[0.2881]	0.0020[0.0012]	0.9067[0.3852]
		θ_{22}	0.3982[0.0464]	0.8996[0.4082]	0.0028[0.0011]	0.9051[0.4981]
		θ_1	0.0928[0.0148]	0.9138[0.1926]	0.0037[0.0009]	0.9185[0.1739]
		θ_2	0.2591[0.0751]	0.9141[0.2428]	0.0032[0.0009]	0.9159[0.2117]
		λ	2.9682[7.6851]	0.9093[0.9898]	0.7582[1.4824]	0.9227[0.3728]
		$R(t)$	0.1628[0.0359]	0.9287[0.0074]	0.4758[0.2718]	0.9325[0.0036]
		$h(t)$	0.2858[0.0769]	0.9259[0.0291]	0.2943[0.1985]	0.9372[0.0045]
		Md(t)	3.74E3[6.12E4]	0.9263[0.0392]	12.0458[3.40E2]	0.9331[0.0062]
100	50	θ_{11}	0.0924[0.0082]	0.9108[0.0864]	0.0028[0.0018]	0.9244[0.0761]
		θ_{12}	0.1592[0.0367]	0.9121[0.1382]	0.0034[0.0017]	0.9382[0.5492]
		θ_{21}	0.0478[0.0028]	0.9013[0.1122]	0.0039[0.0016]	0.9026[0.4266]
		θ_{22}	0.2015[0.0482]	0.9067[0.1989]	0.0027[0.0011]	0.9049[0.3782]
		θ_1	0.1371[0.0244]	0.9256[0.0743]	0.0053[0.0009]	0.9125[0.3345]
		θ_2	0.2897[0.0975]	0.9285[0.1125]	0.0049[0.0008]	0.9227[0.4762]
		λ	2.6584[6.7715]	0.9145[0.8758]	0.7682[1.1425]	0.9234[0.6658]
		$R(t)$	0.1691[0.0372]	0.9362[0.0057]	0.5563[0.3351]	0.9365[0.0048]
		$h(t)$	0.2906[0.0891]	0.9329[0.0291]	0.2293[0.1398]	0.9341[0.0053]
		Md(t)	1.72E3[1.49E5]	0.9318[0.0265]	16.7182[4.54E2]	0.9234[0.0068]
	70	θ_{11}	0.0836[0.0248]	0.9253[0.0349]	0.0014[0.0009]	0.9324[0.0597]
		θ_{12}	0.1592[0.0231]	0.9211[0.0627]	0.0026[0.0008]	0.9317[0.0599]
		θ_{21}	0.1289[0.0026]	0.9109[0.0562]	0.0027[0.0010]	0.9128[0.0482]
		θ_{22}	0.3372[0.0560]	0.9142[0.0799]	0.0013[0.0008]	0.9144[0.0423]
		θ_1	0.0674[0.0291]	0.9328[0.0396]	0.0022[0.0006]	0.9251[0.1818]
		θ_2	0.1993[0.0637]	0.9367[0.0481]	0.0020[0.0005]	0.9353[0.2466]
		λ	0.9994[0.9892]	0.9271[0.6583]	0.0037[0.0007]	0.9365[0.2928]
		$R(t)$	0.1517[0.0359]	0.9465[0.0042]	0.3395[0.0048]	0.9401[0.0032]
		$h(t)$	0.2749[0.0843]	0.9418[0.0286]	0.2268[0.0516]	0.9428[0.0047]
		Md(t)	1.25E3[3.68E4]	0.9424[0.0118]	8.7346[2.96E2]	0.9436[0.0035]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

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TABLE 7 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.15, 0.45, 0.5)$ with $k = 3$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.1379[0.0217]	0.9001[0.7765]	0.0043[0.0026]	0.9145[0.1693]
		θ_{12}	0.3298[0.1672]	0.8814[1.5324]	0.0054[0.0027]	0.9082[0.5891]
		θ_{21}	0.1495[0.0328]	0.8728[1.8251]	0.0042[0.0024]	0.9001[0.5534]
		θ_{22}	0.2824[0.0839]	0.8741[2.8205]	0.0063[0.0018]	0.9003[0.6192]
		θ_{31}	0.1647[0.0314]	0.8725[1.5732]	0.0057[0.0026]	0.9009[0.5621]
		θ_{32}	0.2728[0.0892]	0.8766[1.4083]	0.0061[0.0027]	0.9001[0.5363]
		θ_1	0.3694[0.2675]	0.9143[0.6569]	0.0063[0.0018]	0.9107[0.4814]
		θ_2	0.3739[0.3548]	0.8725[1.0152]	0.0069[0.0049]	0.9048[0.7008]
		λ	3.9563[14.5478]	0.8728[1.6127]	1.9986[2.9125]	0.9007[0.8846]
		$R(t)$	0.1926[0.0488]	0.9084[0.0145]	0.7928[0.5955]	0.9251[0.0082]
		$h(t)$	0.3028[0.0923]	0.9003[0.0591]	0.6761[0.4496]	0.9162[0.0099]
		Md(t)	8.01E3[6.44E5]	0.9005[0.0572]	26.6364[5.62E2]	0.9153[0.0192]
	40	θ_{11}	0.1215[0.0184]	0.9018[0.1824]	0.0028[0.0015]	0.9108[0.0826]
		θ_{12}	0.3082[0.1053]	0.9063[0.3018]	0.0031[0.0010]	0.9392[0.5028]
		θ_{21}	0.1376[0.0326]	0.9005[0.3059]	0.0029[0.0018]	0.9002[0.3743]
		θ_{22}	0.2649[0.0612]	0.8994[0.5236]	0.0052[0.0023]	0.9016[0.4972]
		θ_{31}	0.1468[0.0381]	0.9009[0.3537]	0.0034[0.0012]	0.9009[0.2596]
		θ_{32}	0.2714[0.0694]	0.9001[0.4103]	0.0041[0.0016]	0.9041[0.3297]
		θ_1	0.1053[0.0247]	0.9215[0.1572]	0.0048[0.0013]	0.9154[0.2295]
		θ_2	0.2377[0.0718]	0.9094[0.2874]	0.0039[0.0011]	0.9108[0.5165]
		λ	3.0935[7.9581]	0.8992[0.9992]	0.6967[1.1663]	0.9163[0.3743]
		$R(t)$	0.1722[0.0437]	0.9036[0.0174]	0.4361[0.2795]	0.9251[0.0048]
		$h(t)$	0.2849[0.0853]	0.9182[0.0428]	0.2991[0.2011]	0.9320[0.0060]
		Md(t)	1.83E3[6.15E4]	0.9025[0.0524]	12.1548[3.10E2]	0.9287[0.0081]
100	50	θ_{11}	0.1045[0.0099]	0.9062[0.0891]	0.0031[0.0022]	0.9143[0.0829]
		θ_{12}	0.1671[0.0442]	0.9084[0.1423]	0.0034[0.0019]	0.9357[0.5961]
		θ_{21}	0.0406[0.0023]	0.9033[0.1485]	0.0047[0.0014]	0.9005[0.4582]
		θ_{22}	0.2024[0.0571]	0.9001[0.2273]	0.0035[0.0012]	0.9001[0.4496]
		θ_{31}	0.0735[0.0075]	0.9009[0.1821]	0.0032[0.0014]	0.9002[0.4824]
		θ_{32}	0.1829[0.0408]	0.9003[0.1675]	0.0037[0.0016]	0.9002[0.4345]
		θ_1	0.1430[0.0162]	0.9170[0.0866]	0.0061[0.0010]	0.9158[0.3398]
		θ_2	0.2997[0.0250]	0.9169[0.1382]	0.0050[0.0015]	0.9153[0.2428]
		λ	2.9432[6.9583]	0.9042[0.8825]	0.7729[1.0975]	0.9191[0.6843]
		$R(t)$	0.1850[0.0482]	0.9290[0.0069]	0.5688[0.3736]	0.9236[0.0052]
		$h(t)$	0.2961[0.0904]	0.9127[0.0358]	0.2263[0.1495]	0.9377[0.0059]
		Md(t)	1.79E3[1.46E5]	0.9213[0.0392]	16.7620[4.40E2]	0.9219[0.0062]
	70	θ_{11}	0.0866[0.0208]	0.9143[0.0349]	0.0014[0.0009]	0.9258[0.0632]
		θ_{12}	0.1632[0.0352]	0.9152[0.0705]	0.0030[0.0012]	0.9392[0.0674]
		θ_{21}	0.1326[0.0024]	0.9079[0.0671]	0.0026[0.0009]	0.9124[0.0528]
		θ_{22}	0.3581[0.0642]	0.9135[0.0844]	0.0015[0.0008]	0.9393[0.0517]
		θ_{31}	0.1574[0.0039]	0.9104[0.0682]	0.0036[0.0015]	0.9147[0.0528]
		θ_{32}	0.2162[0.0217]	0.9096[0.0793]	0.0029[0.0014]	0.9135[0.0592]
		θ_1	0.0657[0.0213]	0.9271[0.0476]	0.0034[0.0009]	0.9206[0.1934]
		θ_2	0.2053[0.0125]	0.9218[0.0528]	0.0023[0.0007]	0.9213[0.2628]
		λ	1.1253[0.9960]	0.9235[0.6587]	0.0924[0.0011]	0.9290[0.3082]
		$R(t)$	0.1690[0.0345]	0.9357[0.0053]	0.3491[0.0051]	0.9471[0.0036]

(Continues)

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TABLE 7 (Continued)

n	*m*	parameters	MLE	ACI	Bayes	HPD
		$h(t)$	0.2836[0.0881]	0.9386[0.0355]	0.2025[0.0463]	0.9328[0.0069]
		Md(t)	1.32E3[1.20E4]	0.9394[0.0241]	9.5273[2.51E2]	0.9342[0.0049]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

2.2 | Likelihood function

From (4), the likelihood function of θ_{i1}, θ_{i2} and λ with data $\mathcal{D}_i, 1, 2, \ldots, k$ under i-th group is given by

$$L_i(\theta_{i1}, \theta_{i2}, \lambda; \mathcal{D}_i) \propto \prod_{j=1}^{m_i} \Big[\{ f(x_{ij}; \theta_{i1}, \lambda)[1 - F(x_{ij}; \theta_{i2}, \lambda)] \}^{1_{(\delta_{ij}=1)}} \times \{ f(x_{ij}; \theta_{i2}, \lambda)[1 - S(x_{ij}; \theta_{i1}, \lambda)] \}^{1_{(\delta_{ij}=2)}} \times [S(t; \theta_{i1} + \theta_{i2}, \lambda)]^{r_{ij}} \Big]. \tag{8}$$

Further, let $\mathcal{D} = (\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_k)$, $\Theta = (\theta_{11}, \theta_{12}, \ldots, \theta_{k1}, \theta_{k2})$ and $e_{is} = \sum_{j=1}^{m_i} 1_{(\delta_{ij}=s)}$, $s = 1, 2$, the full likelihood function of Θ and λ is

$$\begin{aligned} L(\Theta, \lambda; \mathcal{D}) &= \prod_{i=1}^{k} L_i(\theta_{i1}, \theta_{i2}, \lambda; \mathcal{D}_i) \\ &= \lambda^{\sum_{i=1}^{k} m_i} \prod_{i=1}^{k} \theta_{i1}^{e_{i1}} \theta_{i2}^{e_{i2}} \times \prod_{i=1}^{k} \prod_{j=1}^{m_i} \frac{e^{-\lambda/x_{ij}}}{1 - e^{-\lambda/x_{ij}}} \\ &\quad \times \prod_{i=1}^{k} \prod_{j=1}^{m_i} [1 - e^{-\lambda/x_{ij}}]^{(r_{ij}+1)[\theta_{i1}+\theta_{i2}]}. \end{aligned} \tag{9}$$

In addition, since the DDF cannot be obtained in explicit form that is equivalent to the difference in competing risks parameters θ_{i1} and θ_{i2}, then the estimation of the DDF is transformed to estimate the differences in θ_{i1} and θ_{i2}. Further, under the assumption that the DDF cannot be neglected, the competing risk parameters θ_1 and θ_2 could be obtained by weighted estimation of θ_{i1} and θ_{i2} respectively, that is, $\theta_1 = \sum_{i=1}^{k} \nu_{i1}\theta_{i1}$ and $\theta_2 = \sum_{i=1}^{k} \nu_{i2}\theta_{i2}$ where $\upsilon_{is}, i = 1, 2, \ldots, k; s = 1, 2$ is the weight coefficient that will be proposed in following contents.

3 | CLASSICAL ESTIMATION

In this section, maximum likelihood estimators (MLEs) of model parameters, reliability indices as well as DDF are established, and approximate confidence intervals (ACIs) are constructed in consequence.

3.1 | Maximum likelihood estimation

From (9), the log-likelihood function of Θ and λ can be rewritten as

$$\begin{aligned} \ell(\Theta, \lambda; \mathcal{D}) &= \sum_{i=1}^{k} m_i \ln \lambda + \sum_{i=1}^{k} e_{i1} \ln \theta_{i1} + \sum_{i=1}^{k} e_{i2} \ln \theta_{i2} + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \ln \left[\frac{e^{-\lambda/x_{ij}}}{1 - e^{-\lambda/x_{ij}}} \right] \\ &\quad + \sum_{i=1}^{k} \theta_{i1} \omega_i(\lambda) + \sum_{i=1}^{k} \theta_{i2} \omega_i(\lambda), \end{aligned} \tag{10}$$

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TABLE 8 Absolute deviations(mean square errors) and coverage probabilities(average lengths) for inverted exponentiated exponential distribution competing risks parameters $(\theta_1, \theta_1, \lambda) = (0.3, 0.6, 2)$ with $k = 3$.

n	m	parameters	MLE	ACI	Bayes	HPD
60	20	θ_{11}	0.1367[0.0224]	0.8739[0.8429]	0.0057[0.0034]	0.9003[0.1724]
		θ_{12}	0.4277[0.0548]	0.8843[1.2688]	0.0062[0.0043]	0.9016[0.7948]
		θ_{21}	0.2296[0.0392]	0.8654[1.6253]	0.0060[0.0031]	0.8728[0.6368]
		θ_{22}	0.5463[0.0762]	0.8521[2.5394]	0.0061[0.0043]	0.8631[0.7272]
		θ_{31}	0.3593[0.0581]	0.8832[1.3620]	0.0053[0.0049]	0.8994[0.6517]
		θ_{32}	0.4628[0.0672]	0.8863[1.6105]	0.0057[0.0045]	0.9003[0.6945]
		θ_1	0.1794[0.0976]	0.8920[0.5423]	0.0071[0.0022]	0.9172[0.4258]
		θ_2	0.4325[0.0517]	0.9003[0.6387]	0.0079[0.0036]	0.9148[0.4893]
		λ	3.9326[15.4271]	0.8628[1.3482]	2.5428[3.2471]	0.9007[0.8999]
		$R(t)$	0.1925[0.0572]	0.9053[0.0094]	0.8263[0.5973]	0.9123[0.0094]
		$h(t)$	0.3327[0.0935]	0.9048[0.0436]	0.7384[0.4943]	0.9127[0.0112]
		Md(t)	8.51E3[6.42E5]	0.9036[0.0528]	32.7468[6.11E2]	0.9136[0.0089]
	40	θ_{11}	0.0955[0.0097]	0.9086[0.0936]	0.0023[0.0019]	0.9134[0.0842]
		θ_{12}	0.1632[0.0381]	0.9071[0.1563]	0.0036[0.0021]	0.9249[0.4638]
		θ_{21}	0.0563[0.0029]	0.9003[0.1426]	0.0041[0.0019]	0.9093[0.0643]
		θ_{22}	0.2160[0.0505]	0.9001[0.2194]	0.0031[0.0014]	0.9001[0.4368]
		θ_{31}	0.0723[0.0063]	0.9003[0.1326]	0.0035[0.0018]	0.9104[0.0576]
		θ_{32}	0.1826[0.0495]	0.9032[0.2473]	0.0036[0.0017]	0.8921[0.5139]
		θ_1	0.0984[0.0186]	0.9058[0.1623]	0.0043[0.0011]	0.9048[0.3362]
		θ_2	0.2372[0.0711]	0.9025[0.2563]	0.0046[0.0010]	0.9035[0.5629]
		λ	3.9216[8.8243]	0.9003[1.0671]	0.7326[1.3785]	0.9128[0.4793]
		$R(t)$	0.1728[0.0436]	0.9130[0.0090]	0.5364[0.3631]	0.9127[0.0088]
		$h(t)$	0.3680[0.0945]	0.9134[0.0288]	0.2902[0.1982]	0.9273[0.0069]
		Md(t)	1.48E3[6.07E4]	0.9155[0.0426]	13.8546[3.02E2]	0.9296[0.0059]
100	50	θ_{11}	0.0944[0.0081]	0.9118[0.0986]	0.0037[0.0016]	0.9219[0.0746]
		θ_{12}	0.1517[0.0329]	0.9173[0.1471]	0.0035[0.0018]	0.9368[0.5471]
		θ_{21}	0.0539[0.0047]	0.8991[0.1497]	0.0039[0.0017]	0.9027[0.1237]
		θ_{22}	0.1986[0.0435]	0.9033[0.2184]	0.0038[0.0016]	0.9033[0.1745]
		θ_{31}	0.0532[0.0042]	0.8993[0.0576]	0.0035[0.0015]	0.9041[0.0289]
		θ_{32}	0.1641[0.0379]	0.8998[0.1754]	0.0039[0.0017]	0.9083[0.3651]
		θ_1	0.1392[0.0236]	0.9162[0.0815]	0.0064[0.0010]	0.9162[0.3628]
		θ_2	0.3285[0.0993]	0.9136[0.1284]	0.0053[0.0016]	0.9140[0.5342]
		λ	3.2341[8.2480]	0.9043[0.8980]	0.8781[1.6078]	0.9124[0.7276]
		$R(t)$	0.1754[0.0492]	0.9214[0.0069]	0.5672[0.3315]	0.9215[0.0050]
		$h(t)$	0.2967[0.0840]	0.9239[0.0284]	0.2468[0.1547]	0.9271[0.0069]
		Md(t)	1.87E3[1.49E4]	0.9261[0.0289]	18.6924[4.61E2]	0.9270[0.0083]
	70	θ_{11}	0.0852[0.0276]	0.9243[0.0475]	0.0017[0.0011]	0.9277[0.0632]
		θ_{12}	0.1425[0.0251]	0.9180[0.0687]	0.0038[0.0014]	0.9328[0.0614]
		θ_{21}	0.1263[0.0027]	0.9136[0.0516]	0.0046[0.0013]	0.9152[0.0584]
		θ_{22}	0.3628[0.0575]	0.9074[0.0835]	0.0018[0.0012]	0.9091[0.0427]
		θ_{31}	0.1534[0.0078]	0.9127[0.0608]	0.0034[0.0008]	0.9169[0.0755]
		θ_{32}	0.3395[0.0263]	0.9038[0.0591]	0.0019[0.0016]	0.9030[0.0429]
		θ_1	0.0619[0.0466]	0.9271[0.0315]	0.0036[0.0007]	0.9292[0.0231]
		θ_2	0.2173[0.0789]	0.9230[0.0419]	0.0039[0.0009]	0.9248[0.0156]
		λ	1.0567[0.9934]	0.9145[0.6632]	0.0025[0.0008]	0.9234[0.3085]
		$R(t)$	0.1730[0.0381]	0.9389[0.0059]	0.3631[0.0056]	0.9392[0.0043]

(Continues)

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TABLE 8 (Continued)

n	*m*	parameters	MLE	ACI	Bayes	HPD
		$h(t)$	0.3275[0.0960]	0.9322[0.0237]	0.2723[0.0693]	0.9393[0.0062]
		Md(t)	1.24E2[3.13E3]	0.9314[0.0114]	9.5829[2.04E2]	0.9381[0.0048]

Abbreviations: ACI, Approximate confidence intervals; HPD, highest posterior density; MLE, maximum likelihood estimator.

TABLE 9 Competing risks data under constant-stress accelerating life testing.

group one	(0.04,1)	(0.10,1)	(0.16,2)	(0.17,2)	(0.42,2)	(0.46,1)	(0.47,1)	(0.47,1)
	(1.05,1)	(1.50,2)	(1.80,2)	(2.23,1)	(2.50,2)	(3.97,2)	(6.10,1)	
group two	(0.07,2)	(0.13,1)	(0.20,2)	(0.29,1)	(0.30,1)	(0.30,2)	(0.30,2)	(0.33,2)
	(0.33,2)	(0.40,1)	(0.40,1)	(0.63,2)	(0.82,1)	(0.84,2)	(1.20,1)	

TABLE 10 Progressive censored competing risks data with designed censoring scenarios using $(n, m) = (15, 12)$.

Data set I: $R = (0,0,0,0,0,0,0,0,0,0,0,3)$						
$(x_{1\cdot}, \delta_{1\cdot})$	(0.04,1)	(0.10,1)	(0.16,2)	(0.17,2)	(0.42,2)	(0.46,1)
	(0.47,1)	(0.47,1)	(1.05,1)	(1.50,2)	(1.80,2)	(2.23,1)
$(x_{2\cdot}, \delta_{2\cdot})$	(0.07,2)	(0.13,1)	(0.20,2)	(0.29,1)	(0.30,1)	(0.30,2)
	(0.30,2)	(0.33,2)	(0.33,2)	(0.40,1)	(0.40,1)	(0.63,2)
Data set II: $R = (0,0,0,1,0,0,0,1,0,0,0,1,0,0,0,1)$						
no.	$(\boldsymbol{s_{11}}, \boldsymbol{s_{12}})$	$(\boldsymbol{x_{1\cdot}}, \boldsymbol{\delta_{1\cdot}})$	$(\boldsymbol{s_{21}}, \boldsymbol{s_{22}})$	$(\boldsymbol{x_{2\cdot}}, \boldsymbol{\delta_{2\cdot}})$		
$(x_{1\cdot}, \delta_{1\cdot})$	(0.04,1)	(0.10,1)	(0.16,2)	(0.17,2)	(0.42,2)	(0.46,1)
	(0.47,1)	(1.05,1)	(1.50,2)	(1.80,2)	(2.50,2)	(3.97,2)
$(x_{2\cdot}, \delta_{2\cdot})$	(0.07,2)	(0.13,1)	(0.20,2)	(0.29,1)	(0.30,1)	(0.30,2)
	(0.30,2)	(0.33,2)	(0.40,1)	(0.63,2)	(0.82,1)	(0.84,2)

where $\omega_i(\lambda) = \sum_{j=1}^{m_i}(r_{ij}+1)\ln[1-e^{-\lambda/x_{ij}}]$.

In consequence, MLEs of $\theta_{i1}, \theta_{i2}, i = 1, 2, \ldots, k$ and λ, say $\hat{\Theta}$ and $\hat{\lambda}$ with $\hat{\Theta} = (\hat{\theta}_{11}, \hat{\theta}_{12}, \ldots, \hat{\theta}_{k1}, \hat{\theta}_{k2})$, are the solutions of following equations as

$$\ell_{\theta_{11}} = 0, \ell_{\theta_{12}} = 0, \ldots, \ell_{\theta_{k1}} = 0, \ell_{\theta_{k2}} = 0 \text{ and } \ell_{\lambda} = 0, \tag{11}$$

where $\ell_{(\cdot)}$ denotes the first derivative of $\ell(\Theta, \lambda; \boldsymbol{D})$ and the details are omitted for concision. Clearly, there is no closed forms of MLEs from (11) and numerical techniques like Newton-Raphson and quasi-Newton methods are needed that sometimes are unstable and may encounter convergence problems. Alternatively, profile likelihood approach is proposed in this paper, where the estimators maximizing the profile likelihood function are also the MLEs obtained from the full likelihood function. For more details on profile likelihood, see Barndorff-Nielsen and Cox[23] for review.

Theorem 1. *Suppose the BPCS competing risks sample (4) follows IEED with parameters θ_1, θ_2 and λ. For $e_{is} > 0, i = 1, 2, \ldots, k; s = 1, 2$, the MLEs of θ_{i1} and θ_{i2} given λ can be expressed as*

$$\dot{\theta}_{i1} = -\frac{e_{i1}}{\omega_i(\lambda)} \quad \textit{and} \quad \dot{\theta}_{i2} = -\frac{e_{i2}}{\omega_i(\lambda)}.$$

Proof. See Appendix A. □

Remark 1. Noted that for $i = 1, 2, \ldots, k$, condition $e_{is} > 0$ in Theorem 1 guarantees the existence of MLE $\dot{\theta}_{is}, s = 1, 2$. Without loss of generality, if $e_{is} = 0$, data $\boldsymbol{D}_i$ under the i-th group does not provide any information about parameter θ_{is}, and then the MLE of θ_{is} does not exist in this case.

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TABLE 11 Point and interval estimates of inverted exponentiated exponential distribution competing risks parameters.

Estimates under Data set I						
	Maximum likelihood method			**Bayesian method**		
parameters	**MLE**	**ESE**	**ACI[length]**	**Bayes**	**ESE**	**HPD[length]**
θ_{11}	0.3180	0.0106	(0.1159,0.5202)[0.4042]	0.3124	3.1E-5	(0.3119,0.3668)[0.0549]
θ_{12}	0.3093	0.0104	(0.1163,0.5184)[0.4021]	0.3113	4.0E-6	(0.3108,0.3594)[0.0468]
θ_{21}	0.4806	0.0080	(0.3049,0.6564)[0.3514]	0.5240	1.8E-3	(0.4806,0.5301)[0.0495]
θ_{22}	0.6729	0.0101	(0.6721,0.6734)[0.0013]	0.6181	3.0E-3	(0.6173,0.7057)[0.0884]
θ_1	0.4106	0.0926	(0.2289,0.5923)[0.3634]	0.3480	3.9E-3	(0.3433,0.3559)[0.0126]
θ_2	7.6148	6.7101	(5.5368,20.7665)[15.2297]	0.3714	0.4668	(0.3669,0.3729)[0.0006]
λ	0.2409	0.0111	(0.2403,0.2414)[0.0011]	0.2933	2.7E-3	(0.2904,0.3060)[0.0156]
$R(t)$	4.2E-6	3.5E-5	(3.9E-6,7.3E-5)[6.9E-5]	0.3733	0.1393	(0.3682,0.3793)[0.0111]
$h(t)$	7.0976	3.9031	(0.5523,14.7476)[14.1953]	0.6190	4.1971	(0.6122,0.6247)[0.0125]
Md(t)	2.5E02	3.9025	(2.4E02,2.6E02)[15.3001]	0.6103	6.4870	(0.6088,0.6183)[0.0095]
Estimates under Data set II						
	Maximum likelihood method			**Bayesian method**		
parameters	**MLE**	**ESE**	**ACI[length]**	**Bayes**	**ESE**	**HPD[length]**
θ_{11}	0.2907	0.1081	(0.0787,0.5027)[0.4240]	0.3093	3.4E-4	(0.2960,0.3097)[0.0137]
θ_{12}	0.4070	0.0600	(0.1796,0.6345)[0.4549]	0.3917	2.3E-4	(0.3760,0.3976)[0.0216]
θ_{21}	0.4873	0.1015	(0.2882,0.6864)[0.3981]	0.4883	1.0E-6	(0.4667,0.5165)[0.0498]
θ_{22}	0.6822	0.0798	(0.5258,0.8386)[0.3128]	0.6197	3.9E-3	(0.6052,0.7049)[0.0997]
θ_1	0.3952	0.1019	(0.1953,0.5950)[0.3997]	0.3563	1.5E-3	(0.3408,0.3632)[0.0224]
θ_2	0.9296	0.5274	(0.1040,1.9633)[1.8593]	0.4515	0.2285	(0.4511,0.4623)[0.0112]
λ	0.2401	0.1078	(0.0288,0.4514)[0.4226]	0.2813	1.7E-3	(0.2802,0.2913)[0.0111]
$S(t)$	0.1292	0.0954	(0.0578,0.3163)[0.2585]	0.3212	0.0368	(0.3203,0.3310)[0.0107]
$h(t)$	1.1721	0.1812	(0.8169,1.5273)[0.7104]	0.6995	0.2233	(0.6145,0.7243)[0.1098]
Md(t)	2.5199	0.1812	(2.1647,2.8752)[0.7105]	0.5099	4.0401	(0.4827,0.5162)[0.0335]

Abbreviations: ESE, estimated standard error; HPD, highest posterior density; MLE, maximum likelihood estimator.

Substituting $\dot{\Theta} = (\dot{\theta}_{11}, \dot{\theta}_{12}, \ldots, \dot{\theta}_{k1}, \dot{\theta}_{k2})$ into (10) and ignoring additive constant terms, the profile log-likelihood function of λ is given by

$$l(\lambda) \propto \sum_{i=1}^{k} m_i \ln \lambda - \sum_{i=1}^{k} e_{i1} \ln [-\omega_i(\lambda)] - \sum_{i=1}^{k} e_{i2} \ln [-\omega_i(\lambda)]$$
$$- \lambda \sum_{i=1}^{k} \sum_{j=1}^{m_i} \frac{1}{x_{ij}} - \sum_{i=1}^{k} \sum_{j=1}^{m_i} \ln[1 - e^{-\lambda/x_{ij}}].$$

Theorem 2. *Suppose the BPCS competing risks sample (4) follows IEED with parameter θ_1, θ_2 and λ. For $e_{is} > 0, i = 1, 2, \ldots, k; s = 1, 2$, the MLE $\hat{\lambda}$ of λ uniquely exists that is the solution of following nonlinear equation*

$$G(\lambda) = 0, \tag{12}$$

where $\omega_i'(\lambda) = \sum_{j=1}^{m_i} (r_{ij} + 1) \frac{\frac{1}{x_{ij}} e^{-\lambda/x_{ij}}}{1 - e^{-\lambda/x_{ij}}}, i = 1, 2, \ldots, k$ *and*

$$G(\lambda) = \frac{\sum_{i=1}^{k} m_i}{\lambda} - \sum_{i=1}^{k} m_i \frac{\omega_i'(\lambda)}{\omega_i(\lambda)} - \sum_{i=1}^{k} \sum_{j=1}^{m_i} \frac{1}{x_{ij}} - \sum_{i=1}^{k} \sum_{j=1}^{m_i} \frac{\frac{1}{x_{ij}} e^{-\lambda/x_{ij}}}{1 - e^{-\lambda/x_{ij}}}.$$

Proof. See Appendix B. □

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Consequently, the MLEs of θ_{i1} and $\theta_{i2}, i = 1, \ldots, k$ are obtained from Theorem 1 as

$$\hat{\theta}_{i1} = -\frac{e_{i1}}{\omega_i(\hat{\lambda})} \text{ and } \hat{\theta}_{i2} = -\frac{e_{i2}}{\omega_i(\hat{\lambda})}, i = 1, 2, \ldots, k.$$

Using invariance principle, reliability indices SF, HRF and MTF of the competing risks IEED can be constructed respectively as

$$\widehat{S}(t) = [1 - e^{-\frac{\hat{\lambda}}{t}}]^{\hat{\theta}_1+\hat{\theta}_2},\ \hat{h}(t) = \frac{(\hat{\theta}_1 + \hat{\theta}_2)\hat{\lambda}e^{-\frac{\hat{\lambda}}{t}}}{t^2(1 - e^{-\frac{\hat{\lambda}}{t}})},\ \widehat{\text{MTF}} = \frac{-\hat{\lambda}}{\ln[1 - (0.5)^{\frac{1}{\hat{\theta}_1+\hat{\theta}_2}}]}. \tag{13}$$

with

$$\hat{\theta}_s = \frac{\sum_{i=1}^{k} \hat{\nu}_{is}\hat{\theta}_{is}}{\sum_{i=1}^{k} \hat{\nu}_{is}} = \frac{\sum_{i=1}^{k} \frac{1}{Var(\hat{\theta}_{is})}\hat{\theta}_{is}}{\sum_{i=1}^{k} \frac{1}{Var(\hat{\theta}_{is})}}, s = 1, 2,$$

where notation $Var(\hat{\theta}_{is})$ is the observed variance of θ_{is} under i-th group which will provided in next subsection.

3.2 | ACIs

The observed Fisher information matrix of (Θ, λ) is constructed as

$$I(\hat{\Theta}, \hat{\lambda}) = \left(\begin{array}{ccccccc} -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{11}\partial\theta_{11}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{11}\partial\theta_{12}} & \cdots & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{11}\partial\theta_{k1}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{11}\partial\theta_{k2}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{11}\partial\lambda} \\ -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{12}\partial\theta_{11}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{12}\partial\theta_{12}} & \cdots & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{12}\partial\theta_{k1}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{12}\partial\theta_{k2}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{12}\partial\lambda} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k1}\partial\theta_{11}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k1}\partial\theta_{12}} & \cdots & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k1}\partial\theta_{k1}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k1}\partial\theta_{k2}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k1}\partial\lambda} \\ -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k2}\partial\theta_{11}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k2}\partial\theta_{12}} & \cdots & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k2}\partial\theta_{k1}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k2}\partial\theta_{k2}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\theta_{k2}\partial\lambda} \\ -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\lambda\partial\theta_{11}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\lambda\partial\theta_{12}} & \cdots & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\lambda\partial\theta_{k1}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\lambda\partial\theta_{k2}} & -\frac{\partial^2 \ell(\Theta,\lambda,D)}{\partial\lambda\partial\lambda} \end{array} \right)\Bigg|_{(\Theta,\lambda)=(\hat{\Theta},\hat{\lambda})},$$

where the second derivatives can be obtained directly and details are omitted for saving space.

Under mild regularity conditions, asymptotic distribution of MLE $(\hat{\Theta}, \hat{\lambda})$ is $(\hat{\Theta}, \hat{\lambda}) - (\Theta, \lambda) \to N(0, I^{-1}((\hat{\Theta}, \hat{\lambda})))$, where $I^{-1}(\hat{\Theta}, \hat{\lambda})$ is the inverse of the observed Fisher information matrix given by

$$I^{-1}(\hat{\Theta}, \hat{\lambda}) = \left(\begin{array}{ccccccc} Var(\hat{\theta}_{11}) & Cov(\hat{\theta}_{11}, \hat{\theta}_{12}) & \cdots & Cov(\hat{\theta}_{11}, \hat{\theta}_{k1}) & Cov(\hat{\theta}_{11}, \hat{\theta}_{k2}) & Cov(\hat{\theta}_{11}, \hat{\lambda}) \\ Cov(\hat{\theta}_{12}, \hat{\theta}_{11}) & Var(\hat{\theta}_{12}) & \cdots & Cov(\hat{\theta}_{12}, \hat{\theta}_{k1}) & Cov(\hat{\theta}_{12}, \hat{\theta}_{k2}) & Cov(\hat{\theta}_{12}, \hat{\lambda}) \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ Cov(\hat{\theta}_{k1}, \hat{\theta}_{11}) & Cov(\hat{\theta}_{k1}, \hat{\theta}_{12}) & \cdots & Var(\hat{\theta}_{k1}) & Cov(\hat{\theta}_{k1}, \hat{\theta}_{k2}) & Cov(\hat{\theta}_{k1}, \hat{\lambda}) \\ Cov(\hat{\theta}_{k2}, \hat{\theta}_{11}) & Cov(\hat{\theta}_{k2}, \hat{\theta}_{12}) & \cdots & Cov(\hat{\theta}_{k2}, \hat{\theta}_{k1}) & Var(\hat{\theta}_{k2}) & Cov(\hat{\theta}_{k2}, \hat{\lambda}) \\ Cov(\hat{\lambda}, \hat{\theta}_{11}) & Cov(\hat{\lambda}, \hat{\theta}_{12}) & \cdots & Cov(\hat{\lambda}, \hat{\theta}_{k1}) & Cov(\hat{\lambda}, \hat{\theta}_{k2}) & Var(\hat{\lambda}) \end{array} \right).$$

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For arbitrary $0 < \gamma < 1$, $100(1-\gamma)\%$ ACI of θ_{is} $(i = 1, 2, \ldots, k; s = 1, 2)$ and λ can be constructed as

$$\left(\hat{\theta}_{is} - z_{\gamma/2}\sqrt{Var(\hat{\theta}_{is})}, \hat{\theta}_{is} + z_{\gamma/2}\sqrt{Var(\hat{\theta}_{is})}\right),$$

and

$$\left(\hat{\lambda} - z_{\gamma/2}\sqrt{Var(\hat{\lambda})}, \hat{\lambda} + z_{\gamma/2}\sqrt{Var(\hat{\lambda})}\right),$$

where z_γ is upper $100\gamma\%$ percentile of standard normal distribution.

Let $\hat{\nu}_{is} = \frac{1}{Var(\hat{\theta}_{is})}$ is MLE of ν_{is} $(i = 1, 2, \ldots, k; s = 1, 2)$. In order to construct ACIs for θ_1, θ_2, reliability indices SF $S(t; \theta_1 + \theta_2, \lambda)$, HRF $h(t; \theta_1 + \theta_2, \lambda)$ and MTF$(\theta_1 + \theta_2, \lambda)$, delta technique and asymptotic distribution theory are used here. Let $g(\Theta, \lambda)$ be an arbitrary function of parameters Θ and λ, the approximate estimate of variance of $g(\Theta, \lambda)$ can be obtained as

$$\widehat{Var}(\hat{g}) = [\nabla g^T(\Theta, \lambda)] I^{-1}(\hat{\Theta}, \hat{\lambda})[\nabla g(\Theta, \lambda)],$$

where

$$\nabla g(\Theta, \lambda) = \left(\frac{\partial g}{\partial \theta_{11}}, \frac{\partial g}{\partial \theta_{12}}, \ldots, \frac{\partial g}{\partial \theta_{k1}}, \frac{\partial g}{\partial \theta_{k2}}, \frac{\partial g}{\partial \lambda}\right)^T \Big|_{(\Theta,\lambda)=(\hat{\Theta},\hat{\lambda})}.$$

Therefore, asymptotic distribution of $\hat{g}(\Theta, \lambda) = g(\hat{\Theta}, \hat{\lambda})$ is given by $\frac{\hat{g}(\Theta,\lambda) - g(\Theta,\lambda)}{\sqrt{\widehat{Var}(\hat{g})}} \to N(0, 1)$, and then the $100(1-\gamma)\%$ ACI of $g(\Theta, \lambda)$ can be constructed as

$$\left(\hat{g}(\Theta, \lambda) - z_{\gamma/2}\sqrt{\widehat{Var}(\hat{g})}, \hat{g}(\Theta, \lambda) + z_{\gamma/2}\sqrt{\widehat{Var}(\hat{g})}\right).$$

Let $g(\Theta, \lambda)$ be associated parameters and reliability indices, the ACIs could be obtained correspondingly and the details are omitted for concision.

4 | BAYESIAN ESTIMATION

In this section, a hierarchical model is conducted for Bayesian estimation. Suppose parameters θ_{i1} and θ_{i2} $(i = 1, \ldots, k)$ follow independent gamma priors as

$$\pi(\theta_{i1}, \theta_{i2}) \propto \theta_{i1}^{a_1 - 1} e^{-b_1 \theta_{i1}} \theta_{i2}^{a_2 - 1} e^{-b_2 \theta_{i2}},$$

implying that the joint prior of Θ is given by

$$\pi(\Theta) \propto \prod_{i=1}^{k} \theta_{i1}^{a_1 - 1} e^{-b_1 \theta_{i1}} \theta_{i2}^{a_2 - 1} e^{-b_2 \theta_{i2}}. \tag{14}$$

The hyper-parameters (a_1, b_1) and (a_2, b_2) in second stage are further assumed to have noninformative priors as

$$\pi(a_1, b_1) \propto 1, a_1 > 0, b_1 > 0 \text{ and } \pi(a_2, b_2) \propto 1, a_2 > 0, b_2 > 0. \tag{15}$$

In addition, an independent noninformative prior for λ is used as

$$\pi(\lambda) \propto 1. \tag{16}$$

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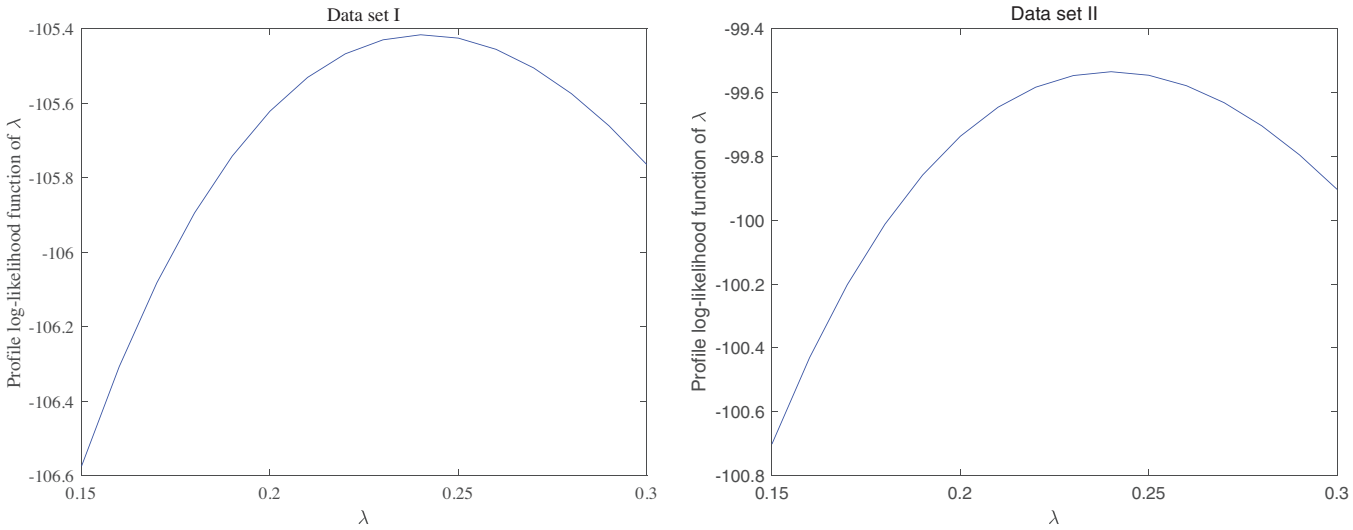


FIGURE 1 Plots of the profile log-likelihood function of λ under Data sets I and II.

Therefore, full posterior density of $\Lambda = (\Theta, a_1, b_1, a_2, b_2, \lambda)$ is given by

$$\begin{aligned}\pi(\Lambda|D) &\propto \pi(\lambda)\pi(a_1, b_1)\pi(a_2, b_2)\pi(\Theta)L(\Theta, \lambda; D)\\ &\propto \lambda^{\sum_{i=1}^{k} m_i} \prod_{i=1}^{k} \theta_{i1}^{a_1+e_{i1}-1}\theta_{i2}^{a_2e_{i2}-1}e^{-b_1\theta_{i1}-b_2\theta_{i2}}\\ &\times \prod_{i=1}^{k}\prod_{j=1}^{m_i} \frac{e^{-\lambda/x_{ij}}}{1-e^{-\lambda/x_{ij}}}[1-e^{-\lambda/x_{ij}}]^{(r_{ij}+1)[\theta_{i1}+\theta_{i2}]}. \end{aligned} \tag{17}$$

For any continuous function $\eta(\Lambda)$ of Λ and under squared error loss, the Bayes estimator of $\eta(\Lambda)$ is given by $\hat{\eta}(\Lambda) = \int \eta(\Lambda)\pi(\Lambda|D)d\Lambda$. Clearly, there is no closed form of $\hat{\eta}(\Lambda)$ and following Monte-Carlo sampling is implemented for complex posterior computation.

4.1 | Posterior analysis and importance sampling

From (17), the conditional posterior density of θ_{is} $(i = 1, 2, \ldots, k; s = 1, 2)$ can be expressed as

$$\pi(\theta_{is}|a_s, b_s, \lambda, D) \propto \theta_{is}^{a_s+e_{is}-1}e^{-b_s\theta_{is}} \times \prod_{j=1}^{m_i}[1-e^{-\lambda/x_{ij}}]^{(r_{ij}+1)\theta_{is}}, \tag{18}$$

the conditional posterior density of a_s and b_s $(s = 1, 2)$ are given by

$$\pi(a_s|\theta_{1s}, \ldots, \theta_{ks}, D) \propto \left(\prod_{i=1}^{k}\theta_{is}\right)^{a_s}, \tag{19}$$

$$\pi(b_s|\theta_{1s}, \ldots, \theta_{ks}, D) \propto \exp\left\{-b_s\sum_{i=1}^{k}\theta_{is}\right\}, \tag{20}$$

and the conditional posterior density of λ are

$$\pi(\lambda|\Theta, D) \propto \lambda^{\sum_{i=1}^{k} m_i} \exp\left\{-\lambda\sum_{i=1}^{k}\sum_{j=1}^{m_i}\frac{1}{x_{ij}}\right\}$$

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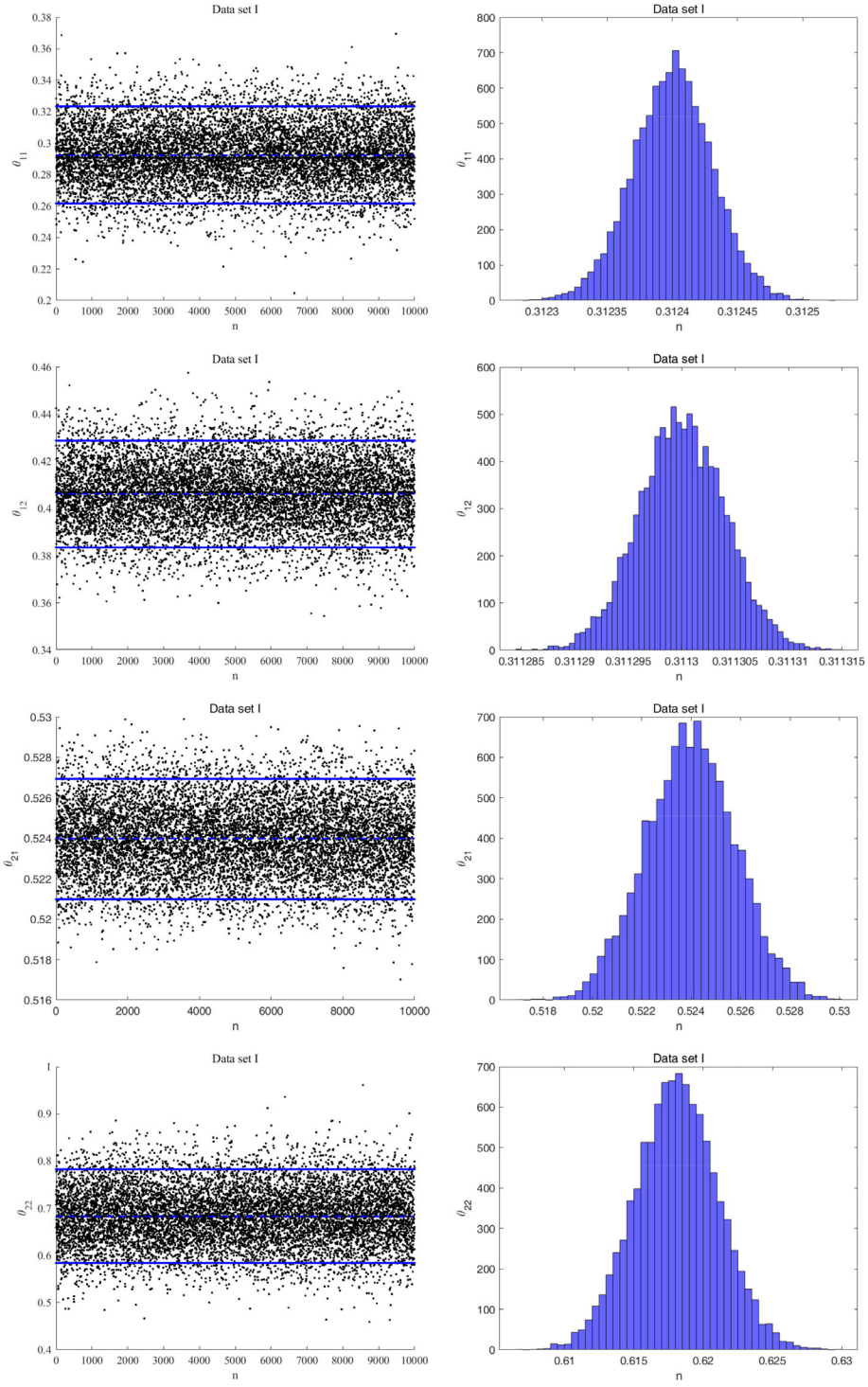


FIGURE 2 Trace plots (the posterior mean (dashed line) with 95% credible intervals (solid lines)) and corresponding histograms of $\theta_{11}, \theta_{12}, \theta_{21}$ and θ_{22} under Data set I.

$$\times \prod_{i=1}^{k} \prod_{j=1}^{m_i} [1 - e^{-\lambda / x_{ij}}]^{(r_{ij}+1)[\theta_{i1}+\theta_{i2}]-1}. \tag{21}$$

It is seen that except for posterior $\pi(b_s|\theta_{1s}, \ldots, \theta_{ks}, \boldsymbol{D})$ in (20) referring to exponential distribution, other conditional posterior densities in (18), (19) and (21) cannot reduce to some well-known models. Therefore, in order to compute Bayes estimates under squared error loss, following Metropolis-Hastings (MH) algorithm is implemented for posterior sampling.

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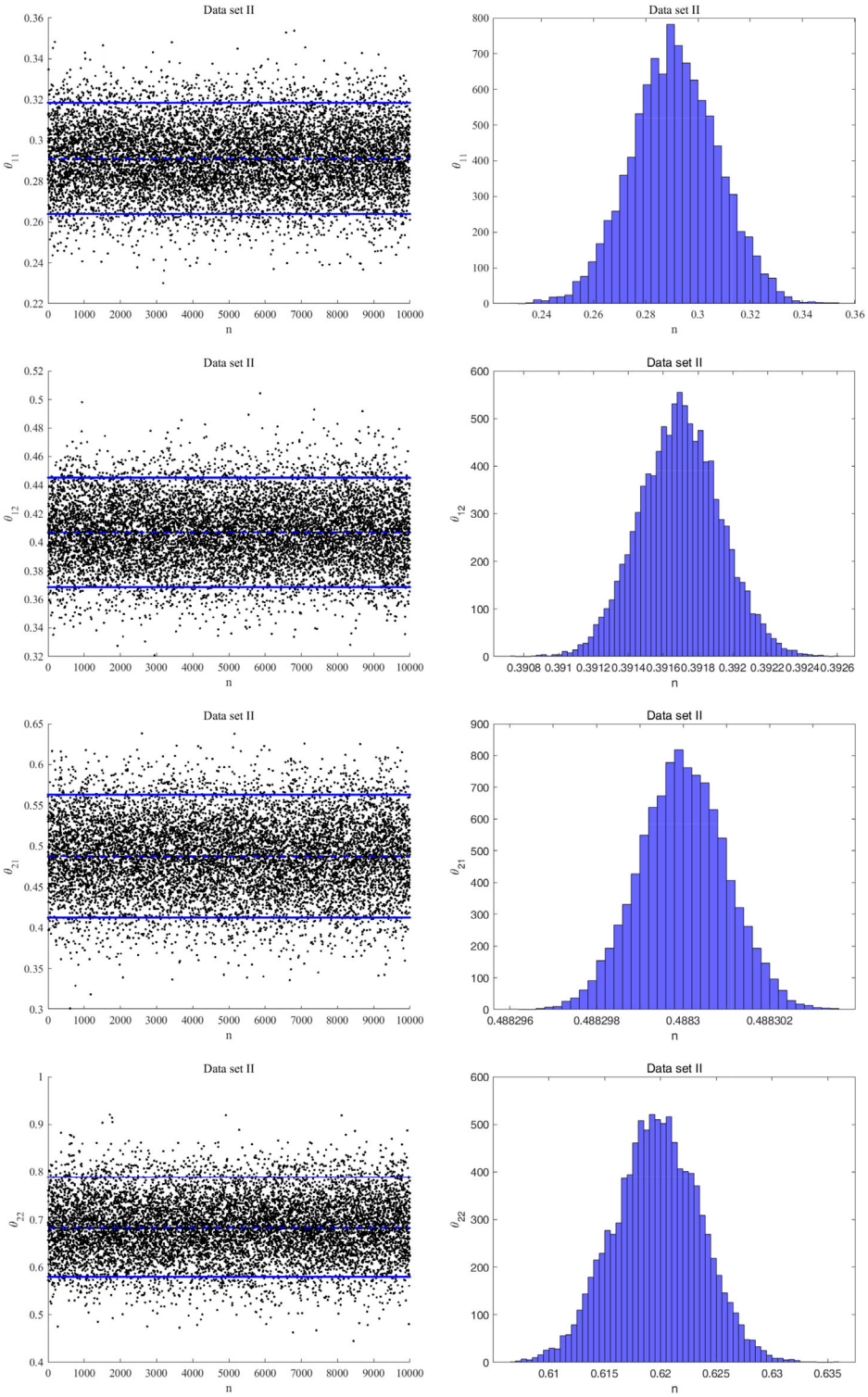


FIGURE 3 Trace plots (the posterior mean [dashed line] with 95% credible intervals [solid lines]) and corresponding histograms of $\theta_{11}, \theta_{12}, \theta_{21}$, and θ_{22} under Data set II.

MH sampling for Bayesian estimation

Step 1 For parameter Λ, set initial value $\Lambda^{(0)} = (\Theta^{(0)}, a_1^{(0)}, b_1^{(0)}, a_2^{(0)}, b_2^{(0)}, \lambda^{(0)})$ with $\Theta^{(0)} = (\theta_{11}^{(0)}, \theta_{12}^{(0)}, \ldots, \theta_{k1}^{(0)}, \theta_{k2}^{(0)})$.
Step 2 Set $d = 1$.

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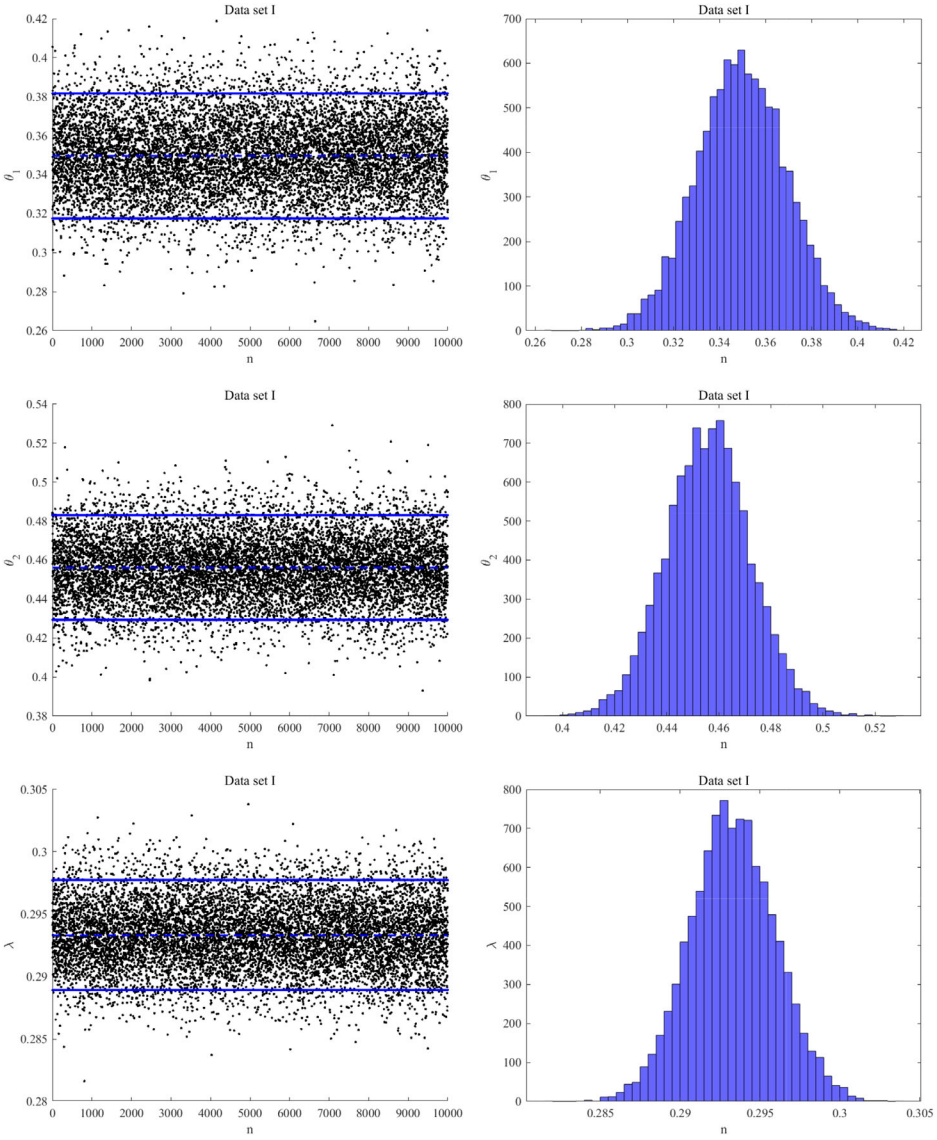


FIGURE 4 Trace plots (the posterior mean [dashed line] with 95% credible intervals [solid lines]) and corresponding histograms of θ_1, θ_2, and λ under Data set I.

Step 3 For given $a_1, b_1, a_2, b_2, \lambda$ and BPCS competing risks data $\mathcal{D}$, generate $\theta_{is}^{(d)}, i = 1, 2, \ldots, k; s = 1, 2$ from $\pi(\theta_{is}^{(d-1)} | a_s^{(d-1)}, b_s^{(d-1)}, \lambda^{(d-1)}, \mathcal{D})$ with normal proposal distribution $N(\theta_{is}^{(d-1)}, V_{\theta_{is}})$, using the following MH algorithm.

(I) generate θ_{is}^* from the proposal distribution $N(\theta_{is}^{(d-1)}, V_{\theta_{is}})$ where $V_{\theta_{is}}$ denotes the variance from the variance-covariance matrix.

(II) calculate the acceptance probability

$$p\left(\theta_{is}^{(d-1)} | \theta_{is}^*\right) = \min\left[1, \frac{\pi(\theta_{is}^* | a_s^{(d-1)}, b_s^{(d-1)}, \lambda^{(d-1)}, \mathcal{D}) Q(\theta_{is}^{(d-1)} | \theta_{is}^*)}{\pi(\theta_{is}^{(d-1)} | a_{is}^{(d-1)}, b_s^{(d-1)}, \lambda^{(d-1)}, data) Q(\theta_{is}^* | \theta_{is}^{(d-1)})}\right],$$

where

$$Q\left(\theta_{is}^{(d-1)} | \theta_{is}^*\right) = \frac{1}{\sqrt{2\pi Var(\theta_{is}^*)}} \exp\left\{-\frac{(\theta_{is}^{(d-1)} - \theta_{is}^*)^2}{2Var(\theta_{is}^*)}\right\},$$

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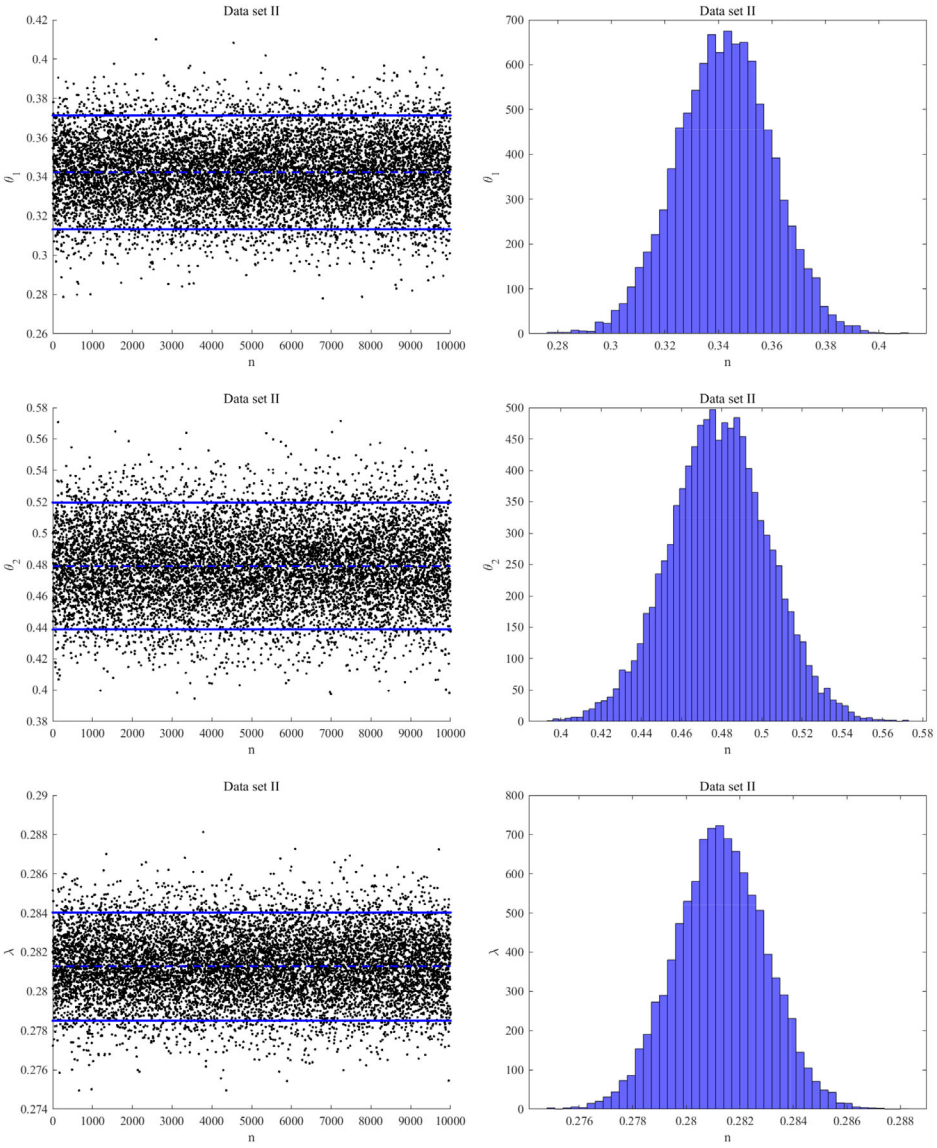


FIGURE 5 Trace plots (the posterior mean [dashed line] with 95% credible intervals [solid lines]) and corresponding histograms of $\theta_1, \theta_2,$ and λ under Data set II.

$$Q\left(\theta_{is}^{*}|\theta_{is}^{(d-1)}\right) = \frac{1}{\sqrt{2\pi Var(\theta_{is}^{(d-1)})}} \exp\left\{-\frac{(\theta_{is}^{*}-\theta_{is}^{(d-1)})^2}{2Var(\theta_{is}^{(d-1)})}\right\}.$$

(III) Generate $u \sim U(0,1)$.

(IV) If $u \leq p(\theta_{is}^{(d-1)}|\theta_{is}^{*})$, acceptance the proposal distribution and set $\theta_{is}^{(d)} = \theta_{is}^{*}$. Otherwise, reject the proposal distribution and set $\theta^{(d)} = \theta^{(d-1)}$.

Step 4 Using MH technique, generate $a_s^{(d)}, s = 1,2$ from $\pi(a_s^{(d-1)}|\theta_{1s}^{(d)}, \ldots, \theta_{ks}^{(d)}, \mathcal{D})$ with normal proposal distribution $N(a_s^{(d-1)}, V_{a_s})$.

Step 5 Generate $b_s^{(d)}, s = 1,2$ from exponential distribution with parameter $\sum_{i=1}^{k} \theta_{is}^{(d)}$.

Step 6 Using MH technique, generate $\lambda^{(d)}$ from $\pi(\lambda^{(d-1)}|\Theta^{(d)}, \mathcal{D})$ with normal proposal distribution $N(\lambda^{(d-1)}, V_\lambda)$, where $\Theta^{(d)} = (\theta_{11}^{(d)}, \theta_{12}^{(d)}, \ldots, \theta_{k1}^{(d)}, \theta_{k2}^{(d)})$.

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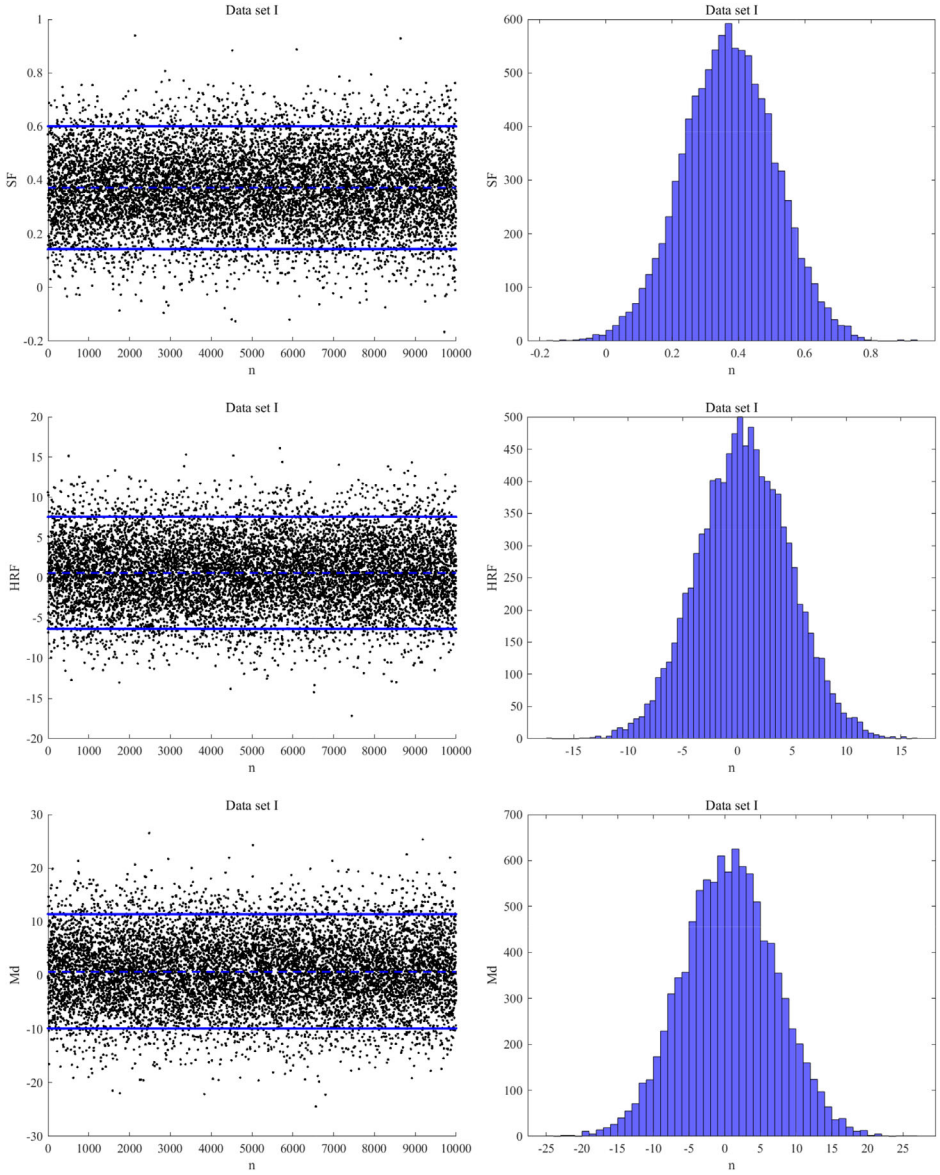


FIGURE 6 Trace plots (the posterior mean [dashed line] with 95% credible intervals [solid lines]) and corresponding histograms of SF, HRF, and Md under Data set I. HRF, hazard rate function; SF, survival function.

Step 7 Compute estimate of $\theta_s, s = 1, 2$ in dth iteration as follows

$$\theta_s^{(d)} = \frac{\sum_{i=1}^{k} \frac{1}{Var(\theta_{is}|D)} \theta_{is}^{(d)}}{\sum_{i=1}^{k} \frac{1}{Var(\theta_{is}|D)}}, s = 1, 2,$$

where $Var(\theta_{is}|D) = \frac{1}{d}\sum_{l=1}^{d}(\theta_{is}^{(l)} - \frac{1}{d}\sum_{e=1}^{d}\theta_{is}^{(e)})^2$.

Step 8 The reliability indices in d-th iteration can be estimated respectively at mission time t as $S(t, \theta_1^{(d)} + \theta_2^{(d)}, \lambda^{(d)}), h(t, \theta_1^{(d)} + \theta_2^{(d)}, \lambda^{(d)})$ and $\mathrm{MTF}(\theta_1^{(d)} + \theta_2^{(d)}, \lambda^{(d)})$ by using substitution method.

Step 9 Set $d = d + 1$.

Step 10 Repeat Steps 3-8 N times, and N random samples of Λ called $\Lambda^{(1)}, \Lambda^{(2)}, \ldots, \Lambda^{(N)}$ can be obtained. Consequently, N estimates of reliability indices $S(t, \theta_1 + \theta_2, \lambda), h(t, \theta_1 + \theta_2, \lambda)$ and $\mathrm{MTF}(\theta_1 + \theta_2, \lambda)$ are also obtained as $S^{(1)}, S^{(2)}, \ldots, S^{(N)}; h^{(1)}, h^{(2)}, \ldots, h^{(N)}$ and $\mathrm{MTF}^{(1)}, \mathrm{MTF}^{(2)}, \ldots, \mathrm{MTF}^{(N)}$, respectively.

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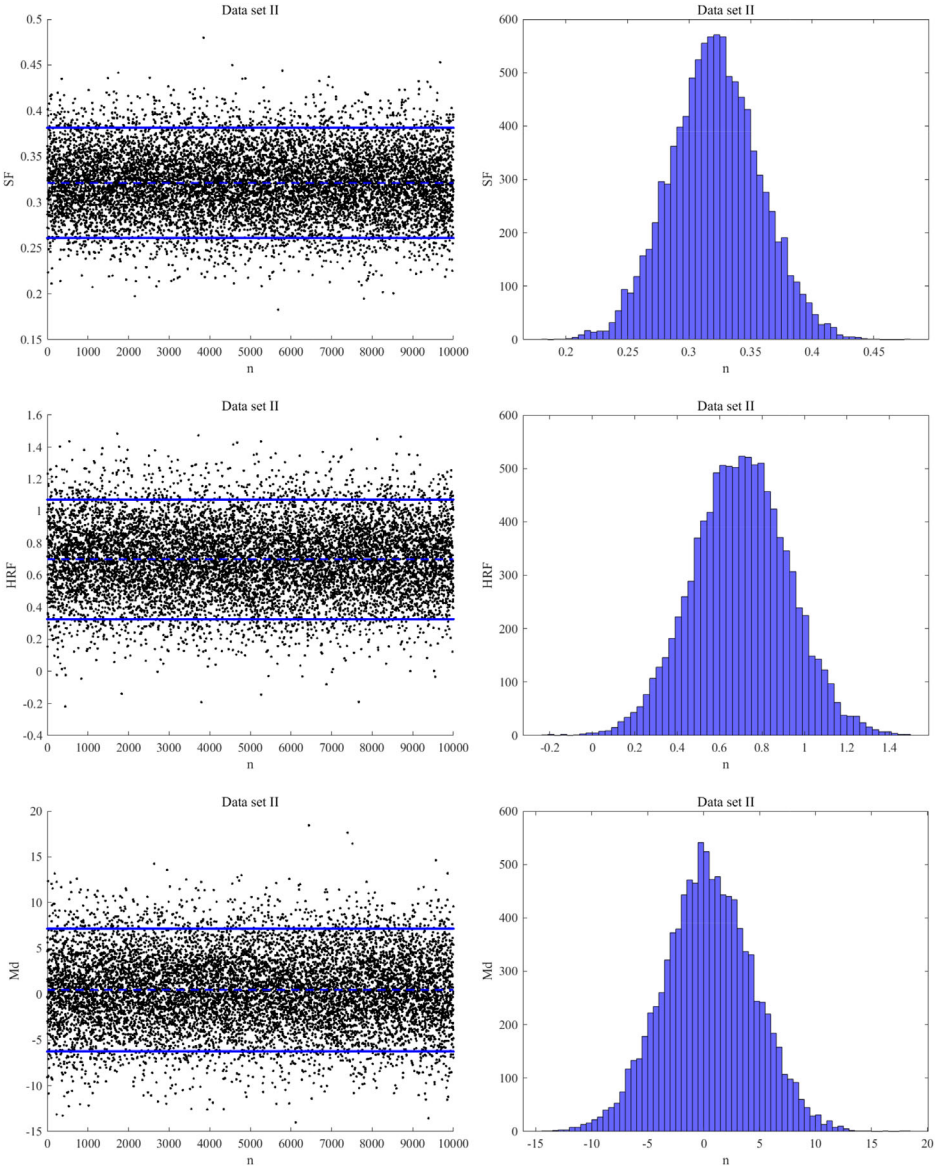


FIGURE 7 Trace plots (the posterior mean [dashed line] with 95% credible intervals [solid lines]) and corresponding histograms of SF, HRF, and Md under Data set II. HRF, hazard rate function; SF, survival function.

Based on above procedure, in order to ensure the convergence of the algorithm and to eliminate the influence of initial value selection, first M simulated samples were discarded. Therefore, the Bayes estimates of parameters $\theta_{11}, \theta_{12}, \ldots, \theta_{k1}, \theta_{k2}$, $\theta_1, \theta_2, \lambda$ and reliability indices can be obtained by the mean of the remaining selected posterior samples. Further, let $\eta^{(i)} = \eta(\Lambda^{(M+i)}), i = 1, 2, \ldots, N - M$, Bayesian highest posterior density (HPD) credible interval of $\eta(\Lambda)$ could be constructed as follow. First arrange all estimates $\eta^{(i)}, i = 1, 2, \ldots, N - M$ in an ascending order as $\eta^{[1]}, \eta^{[2]}, \ldots, \eta^{[N-M]}$. For arbitrary $0 < \gamma < 1$, a series of $100(1 - \gamma)$ credible interval of $\eta(\Lambda)$ can be obtained as

$$(\eta^{[s]}, \eta^{[s+N-M-[\gamma(N-M)+1]]}), s = 1, 2, \ldots, [(N - M)\gamma],$$

where $[y]$ is the greatest integer less than or equal to y. Therefore, $100(1 - \gamma)\%$ HPD credible interval of $\eta(\Lambda)$ can be selected as l^*th one satisfying

$$\eta^{[l^*+N-M-[\gamma(N-M)+1]]} - \eta^{[l^*]} = \min_{l=1}^{(N-M)\gamma} (\eta^{[l+N-M-[\gamma(N-M)+1]]} - \eta^{[l]}).$$

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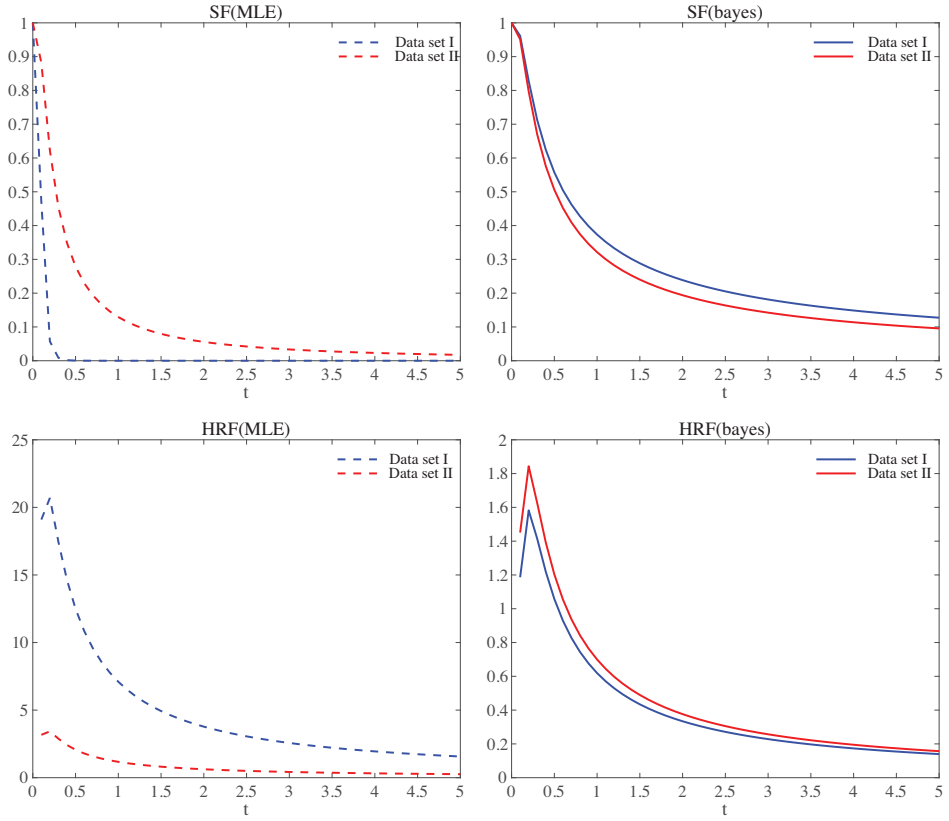


FIGURE 8 Tendencies of SF and HRF along with time based on MLE (dashed lines) and Bayes (solid line) estimates under Data set I and II. HRF, hazard rate function; MLE, maximum likelihood estimators; SF, survival function.

5 | NUMERICAL ILLUSTRATION

5.1 | Simulation studies

In this section, extensive simulation studies are conducted to investigate the performance of different methods. To compare the accuracy of different results, point estimates are evaluated in terms of absolute deviation (AB) and mean square error (MSE), while interval estimates are compared by using average length (AL) and coverage probability (CP). For incorporating test facilities' influence effect, we simulate a random noise ε_i to the data from i-th ($i = 1, 2, \ldots, k$) facility and assume that the noise has normal distribution with $N(0, 0.001)$.

In this paper, various values of parameter $(\theta_1, \theta_2, \lambda)$, sample sizes n, m and the CSs are used for simulation. In addition, sample sizes, CS under different testing facilities are assumed to be same for concision, that is, $n_1 = \cdots = n_k = n$, $m_1 = \cdots = m_k = m$ and $R_1 = \cdots = R_k$. Moreover, an uniformly CS $R_i = (1, \ldots, 1, 0)$ is employed for sake of concision. Therefore, criteria quantities ABs, MSEs, CPs and ALs based on 10,000 repetitions are tabulated in Tables 1–4, where confidence level is 0.95 and estimates of reliability indices are obtained under mission time $t = 1$.

From Tables 1–4, some conclusions are stated as follows. Quantities ABs and MSEs of point estimates decrease when sample size n or censoring number m or their combinations increase in all simulated scenarios, the results indicate that the proposed results are consistent and works satisfactorily in general. Under fixed designed scenarios, the performance of Bayesian estimates of different parameters are superior to traditional MLEs in terms of ABs and MSEs, whereas similar phenomenon also appears for reliability indices under different cases. Similarly, the CPs (ALs) of all interval estimates ACIs and HPDs become larger (smaller) in general when effective sample size (i.e., n or m or their combinations) increases. Comparing with ACI estimates, CPs and ALs of the Bayesian HPD credible intervals come out to be superior that outperform the method of ACIs in general. One could also note that the CPs obtained here fairly attains the prescribed nominal confidence level when sample size gets larger. In addition, for competing risk parameters θ_{is} with $s = 1$ and 2 under different block scenarios, the ALs of their confidence intervals obtained from different groups are not close to each other and

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appear to be different with each other. This phenomenon reflects that the DDF among different blocks indeed exists, and the results show the total facilities effect of DDF in consequence and cannot be negligible.

For the tabulated results shown in Tables 1–4, the simulation studies are conducted based on the assumption that the random noise ε_i for each facility follows identical normal distribution $N(0, 0.001)$. Although the detailed effect among the DDFs of different facilities seems different as mentioned in previous simulated results, another numerical simulation is also conducted to compare and evaluate the performance of DDF more clearly, where different noises with $N(0, 0.005i)$ are incorporated into the i-th facility and other designed scenarios as same as before. Under this assumption, criteria quantities for different estimates are further provided in Tables 5–8. Consequently, one could also observe following conclusions. Firstly, the proposed hierarchical model outperforms likelihood method in general where both point and interval estimates of parameter and reliability performances from Bayesian approach is superior to the classical estimates in terms of criteria quantities, and the finding is consistent with the results presented in Tables 1–4. On the other hand, when general scenarios with different DDF effects are implemented, the difference for given competing risks parameters have raised visibly, and the ALs from both likelihood and Bayesian procedures are still not close to each other, which also evidently indicate the effect of DDF that could be not negligible from practical perspective.

5.2 | Data analysis

A real-life data analysis is presented for illustration purpose. The real data is original provided in Ebrahimi[24] from an accelerating life test and further studied by Bai et al.[25] under a bivariate Weibull model. In this illustration, it is assumed that the DDF are caused by different stress levels in the experiment. The complete failure data of size 15 in two testing groups are presented in Table 9.

From Table 9, two sets of block progressively competing risks data are generated showing in Table 10, where same CSs are used for two groups as shown in data sets I and II for concision. Since there is no extra information for model parameters, all hyper-parameters are taken to be 0.001, which means that priors are proper but almost noninformative. Therefore, point and interval estimates are calculated and the results are presented in Table 11, where the estimated standard error (ESE) for point estimates is obtained for comparing the accuracy of different results, and interval estimates are obtained under 0.95 confidence level.

From Table 11, the MLEs and Bayesian estimates of different parameters and IEED reliability indices perform similarly by comparing the values of their results, whereas the ESE obtained from hierarchical Bayesian model are relatively smaller than those of MLEs in both data sets I and II. In addition, the HPD credible intervals from the hierarchical model along with almost non-informative prior outperform than ACIs in terms of interval lengths in general. Hence, the proposed hierarchical model is suggested to implement when BPCS competing risks, data is available where higher inferential accuracy for both parameters and reliability performances achieves. Moreover, the effect of DDF also appears which affects the inferential results of different estimates and are reflected by the interval lengths of parameters $(\theta_{11}, \theta_{21})$ and $(\theta_{12}, \theta_{22})$, respectively. Further, based the proposed BPCS competing risks data sets I and II, graphs of the profile log-likelihood function of parameter λ are presented in Figure 1. It is conducted that the plots of the profile log-likelihood function is unimodal being consistent with the result given in Theorem 2.

For illustrating the performance of the sampling effect of the hierarchical model, trace plots with their sample means and 95% credible intervals as well as the corresponding histograms of the associated chain values for $\theta_{11}, \theta_{12}, \theta_{21}, \theta_{22}$ and $\theta_1, \theta_2, \lambda$ are presented in Figures 2–7, and the plots indicate a good mixing performance for our proposed methods. In addition, the changing tendencies of SF and HRF of the IEEDs with parameters $\theta_1 + \theta_2$ and λ are also presented in Figure 8 based on two data sets, where the detailed plots of SF and HRF are presented via the MLEs and the Bayesian estimates, respectively. Consequently, it is also noted that SF decreases significantly over time while HRF increases.

6 | CONCLUDING REMARKS

In this paper, inference for competing risks model is studied under block progressive censoring. When the lifetime of units follows IEED and DDF from testing facilities is nonnegligible, classical likelihood estimation is presented for competing risks parameters, HRF, SF, MTF, and DDF, where existence and uniqueness of the MLEs are also established. In addition, Bayesian hierarchical model has proposed, and Monte-Carlo sampling is provided for computation. Extensive simulation studies and the real-life application are conducted, and the results conclude that the proposed hierarchical model seems

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attractive and outperforms classical estimates under BPCS. Although inference is explored for IEED competing risks model under BPCS with DDF effect, the proposed methods could be also implemented when other lifetime distributions like bathtub-shaped Chen, Weibull, Burr XII, Gompertz, Kumaraswamy and others testing schemes like hybrid censoring, progressive first-failure censoring, Type-I/-II censoring with proper modifications. In particular, as a direct application and generalization, the results could be also obtained parallelly for a general class of IEED with CDF

$$F(t;\alpha,\lambda) = 1 - [1 - e^{-\lambda Q(1/t)}]^{\theta},\ t > 0,$$

proposed by Ghitany et al.[26] (see also Kizilaslan,[27] Lodhi et al.,[28] Dey et al.[17]), where $Q(\cdot)$ is baseline function with $Q(0) = 0$ and $Q(\infty) = \infty$. For further studies, it is also interesting to discuss optimal design problems under block testing by taking the effect of the DDF into account which will be reported in future.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ORCID

Liang Wang https://orcid.org/0000-0002-2600-5112
Shuo-Jye Wu https://orcid.org/0000-0001-7294-8018
Yogesh Mani Tripathi https://orcid.org/0000-0002-9687-6036

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APPENDIX A

A. Proof of Theorem 1

By taking derivatives, the MLEs $\dot{\theta}_{i1}$ and $\dot{\theta}_{i2}, i = 1, 2, \ldots, k$ is obtained directly. In following, it is shown that the likelihood function arrives its maximum at $(\dot{\theta}_{11}, \dot{\theta}_{12}, \ldots, \dot{\theta}_{k1}, \dot{\theta}_{k2})$.

Using inequality $\ln t \leq t - 1$ for $t = \frac{\theta_{i1}}{\dot{\theta}_{i1}}(> 0)$ and $\frac{\theta_{i2}}{\dot{\theta}_{i2}}(> 0)$, one has

$$\ln \theta_{i1} \leq= -\frac{\theta_{i1}\omega_i(\lambda)}{e_{i1}} - 1 + \ln \dot{\theta}_{i1} \text{ and } \ln \theta_{i2} \leq -\frac{\theta_{i2}\omega_i(\lambda)}{e_{i2}} - 1 + \ln \dot{\theta}_{i2}.$$

Ignoring the constant terms and using above inequalities as well as the expressions of $(\dot{\theta}_{11}, \dot{\theta}_{12}, \ldots, \dot{\theta}_{k1}, \dot{\theta}_{k2})$, it's noted that

$$\begin{aligned}\ell(\Theta, \lambda; \mathcal{D}) &\leq \sum_{i=1}^{k} m_i \ln \lambda + \sum_{i=1}^{k} e_{i1} \ln \dot{\theta}_{i1} + \sum_{i=1}^{k} e_{i2} \ln \dot{\theta}_{i2} \\ &+ \sum_{i=1}^{k} \sum_{j=1}^{m_i} \ln \left[\frac{e^{-\lambda/x_{ij}}}{1 - e^{-\lambda/x_{ij}}}\right] + \sum_{i=1}^{k} \dot{\theta}_{i1}\omega_i(\lambda) + \sum_{i=1}^{k} \dot{\theta}_{i2}\omega_i(\lambda) \\ &= \ell(\dot{\Theta}, \lambda; \mathcal{D}),\end{aligned}$$

where equality holds if and only if $\Theta = \dot{\Theta} = (\dot{\theta}_{11}, \dot{\theta}_{12}, \ldots, \dot{\theta}_{k1}, \dot{\theta}_{k2})$. Therefore, the assertion is completed.

B. Proof of Theorem 2

The likelihood equation of MLE $\hat{\lambda}$ can be obtained. In following, the existence and uniqueness of $\hat{\lambda}$ is established, respectively.

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Before further proof, some useful results from Ghitany et al.[29] are provided as follows.

(a) $\lim_{z\to 0}(\frac{1}{z}-\frac{1}{1-e^{-z}})=-\frac{1}{2}$;
(b) $\lim_{z\to 0}\frac{z\,e^{-z}}{1-e^{-z}}=1$;
(c) $\lim_{z\to 0} z|\log(1-e^{-z})|=0$;
(d) $\lim_{z\to\infty}\frac{|\log(1-e^{-z})|}{e^{-z}}=1$;
(e) For $z>0$, $z^k e^{-z}<(1-e^{-z})^k$, $k=1,2$.

Using results (a), (b), (c), and (d), one directly has that

$$\lim_{\lambda\to 0} G(\lambda)=+\infty \text{ and } \lim_{\lambda\to\infty} G(\lambda)=\sum_{i=1}^{k}\sum_{j=1}^{m_i}\left[\frac{1}{x_{im_i}}-\frac{1}{x_{ij}}\right]<0.$$

Therefore, function $G(\lambda)$ changes from positive to negative in $(0,\infty)$ and the MLE of λ exists.

By taking derivative, it is noted that

$$\frac{dG(\lambda)}{d\lambda}=-\frac{g_1(\lambda)}{\lambda^2}+\sum_{i=1}^{k}\frac{m_i g_2^i(\lambda)}{[w_i(\lambda)]^2},$$

with $g_1(\lambda)=\sum_{i=1}^{k}\sum_{j=1}^{m_i}(1-\frac{\lambda^2}{x_{ij}^2}\frac{e^{-\lambda/x_{ij}}}{[1-e^{-\lambda/x_{ij}}]^2})$ and

$$g_2^i(\lambda)=\left(\sum_{j=1}^{m_i}\frac{(r_{ij}+1)}{x_{ij}^2}\frac{e^{-\lambda/x_{ij}}}{[1-e^{-\lambda/x_{ij}}]^2}\right)\cdot\left(\sum_{j=1}^{m_i}(r_{ij}+1)\ln[1-e^{-\lambda/x_{ij}}]\right)$$
$$+\left(\sum_{j=1}^{m_i}\frac{(r_{ij}+1)}{x_{ij}}\frac{e^{-\lambda/x_{ij}}}{1-e^{-\lambda/x_{ij}}}\right)^2, i=1,2\ldots,k.$$

Using result (e) with $k=2$, one has $g_1(\lambda)>0$. Further, using inequality $\ln(1-e^{-z})<-e^{-z}, z>0$ and Cauchy-Schwarz inequality, it is seen that $g_2^i(\lambda)\le 0$ for $i=1,2\ldots,k$ implying that $G(\lambda)$ decreases in λ. Therefore, equation $G(\lambda)=0$ has a unique root and the assertion is completed.

AUTHOR BIOGRAPHIES

Liang Wang received the PhD degree in Applied Mathematics from Northwestern Polytechnical University, Xi'an, China, in 2012. He is currently a full Professor with the School of Mathematics, Yunnan Normal University, Kunming, China. His research interests include applied probability and statistics, reliability analysis and life testing.

Shuo-Jye Wu is a professor in the Department of Statistics at Tamkang University. He received his PhD in statistics from the University of Wisconsin-Madison. His professional interests are in the development and application of statistical methodology for problems in reliability.

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Huizhong Lin received her Bachelor degree in Statistics in the Department of Mathematics and Statistics at Jishou University at 2020. She now is master student in statistics at School of Mathematics of Yunnan Normal University. Her research interests are applied statistics and reliability analysis.

Yogesh Mani Tripathi received his PhD degree from the Department of Mathematics, Indian Institute of Technology Kharagpur, India, under the guidance of Prof. Somesh Kumar. He was a Postdoctoral Fellow with Prof. Eric Marchand with the Department of Mathematics at University of Sherbrooke, Canada and with G. S. Shieh at Institute of Statistical Science, Academia Sinica, Taiwan. Currently, he is an Associate Professor with the Department of Mathematics, Indian Institute of Technology Patna, India. His research interests are in decision theory, life-testing and reliability analysis.

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